

Jordan Algebra Formalization in Lean 4

A Mathematical Rendering

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Chapter 1

Introduction

This document presents a LaTeX rendering of the Jordan algebra formalization from the `af-tests` Lean 4 project. Each section corresponds to a Lean source file, presenting the formal definitions and theorems in mathematical notation.

Notation:

- $a \circ b$ denotes the Jordan product of a and b
- a^2 denotes $a \circ a$ (Jordan square)
- L_a denotes the left multiplication operator $x \mapsto a \circ x$
- U_a denotes the quadratic operator $U_a(x) = 2L_a^2(x) - L_{a^2}(x)$
- $P_\lambda(e)$ denotes the Peirce λ -eigenspace for idempotent e

Chapter 2

Core Definitions

2.1 Basic

Definition 2.1.1 (JordanAlgebra). A **Jordan algebra** over $\mathbb{R}$ is a type J with an additive commutative group structure and an $\mathbb{R}$ -module structure, equipped with:

- A bilinear product $\circ : J \times J \rightarrow J$
- An identity element $\mathbf{1} \in J$

satisfying:

1. (Commutativity) $\forall a, b \in J: a \circ b = b \circ a$
2. (Jordan identity) $\forall a, b \in J: (a \circ b) \circ (a \circ a) = a \circ (b \circ (a \circ a))$
3. (Left identity) $\forall a \in J: \mathbf{1} \circ a = a$
4. (Left distributivity) $\forall a, b, c \in J: a \circ (b + c) = (a \circ b) + (a \circ c)$
5. (Scalar compatibility) $\forall r \in \mathbb{R}, \forall a, b \in J: (r \cdot a) \circ b = r \cdot (a \circ b)$

2.1.1 Basic Properties

Lemma 2.1.2 (jmul_jone). For all $a \in J: a \circ \mathbf{1} = a$.

Lemma 2.1.3 (add_jmul). For all $a, b, c \in J: (a + b) \circ c = (a \circ c) + (b \circ c)$.

Lemma 2.1.4 (smul_jmul). For all $r \in \mathbb{R}$ and $a, b \in J: a \circ (r \cdot b) = r \cdot (a \circ b)$.

Lemma 2.1.5 (jmul_zero). For all $a \in J: a \circ 0 = 0$.

Lemma 2.1.6 (zero_jmul). For all $a \in J: 0 \circ a = 0$.

Lemma 2.1.7 (jmul_neg). For all $a, b \in J: a \circ (-b) = -(a \circ b)$.

Lemma 2.1.8 (neg_jmul). For all $a, b \in J: (-a) \circ b = -(a \circ b)$.

Lemma 2.1.9 (jmul_sub). For all $a, b, c \in J: a \circ (b - c) = (a \circ b) - (a \circ c)$.

Lemma 2.1.10 (sub_jmul). For all $a, b, c \in J: (a - b) \circ c = (a \circ c) - (b \circ c)$.

2.1.2 Square and Powers

Definition 2.1.11 (jsq). The **Jordan square** of an element $a \in J$ is:

$$a^2 := a \circ a$$

Lemma 2.1.12 (jsq_def). For all $a \in J$: $a^2 = a \circ a$.

Theorem 2.1.13 (jordan_identity'). For all $a, b \in J$:

$$(a \circ b) \circ a^2 = a \circ (b \circ a^2)$$

Definition 2.1.14 (jpow). The **Jordan power** of an element $a \in J$ is defined recursively:

$$\begin{aligned} a^0 &:= \mathbf{1} \\ a^{n+1} &:= a \circ a^n \end{aligned}$$

Lemma 2.1.15 (jpow_zero). For all $a \in J$: $a^0 = \mathbf{1}$.

Lemma 2.1.16 (jpow_succ). For all $a \in J$ and $n \in \mathbb{N}$: $a^{n+1} = a \circ a^n$.

Lemma 2.1.17 (jpow_one). For all $a \in J$: $a^1 = a$.

Lemma 2.1.18 (jpow_two). For all $a \in J$: $a^2 = a^2$.

2.1.3 Idempotents

Definition 2.1.19 (IsIdempotent). An element $e \in J$ is **idempotent** if:

$$e^2 = e$$

Lemma 2.1.20 (IsIdempotent.jsq_eq_self). If $e \in J$ is idempotent, then $e^2 = e$.

Theorem 2.1.21 (isIdempotent_zero). The zero element $0 \in J$ is idempotent.

Theorem 2.1.22 (isIdempotent_jone). The identity element $\mathbf{1} \in J$ is idempotent.

Theorem 2.1.23 (IsIdempotent.complement). If $e \in J$ is idempotent, then $\mathbf{1} - e$ is also idempotent.

2.2 Product

2.2.1 Left Multiplication Operator

Definition 2.2.1 (L). The **left multiplication operator** by $a \in J$ is the $\mathbb{R}$ -linear map $L_a : J \rightarrow J$ defined by:

$$L_a(b) := a \circ b$$

with linearity properties:

- $L_a(b + c) = a \circ b + a \circ c$
- $L_a(r \cdot b) = r \cdot (a \circ b)$

Lemma 2.2.2 (L_apply). For all $a, b \in J$:

$$L_a(b) = a \circ b$$

Theorem 2.2.3 (L_one). The left multiplication operator by the identity is the identity map:

$$L_1 = \text{id}$$

Lemma 2.2.4 (L_comm). For all $a \in J$:

$$L_a = L_a$$

Definition 2.2.5 (Lsq). The **squaring map** $\text{Lsq} : J \rightarrow J$ is defined by:

$$\text{Lsq}(a) := a^2$$

Lemma 2.2.6 (Lsq_eq_L_apply). For all $a \in J$:

$$\text{Lsq}(a) = L_a(a)$$

2.2.2 Jordan Identity for Operators

Theorem 2.2.7 (L_comm_L_sq). For all $a \in J$, the operators L_a and L_{a^2} commute:

$$L_a \circ L_{a^2} = L_{a^2} \circ L_a$$

This is the operator form of the Jordan identity.

Corollary 2.2.8 (L_commute_L_sq). For all $a \in J$:

$$\text{Commute}(L_a, L_{a^2})$$

2.2.3 Power Properties

Lemma 2.2.9 (jpow_add_one). For all $a \in J$ and $n \in \mathbb{N}$:

$$a^{n+1} = a \circ a^n$$

Lemma 2.2.10 (L_jpow_succ). For all $a \in J$ and $n \in \mathbb{N}$:

$$a^{n+1} = L_a(a^n)$$

Lemma 2.2.11 (jmul_jpow). For all $a \in J$ and $n \in \mathbb{N}$, powers associate:

$$a \circ a^n = a^{n+1}$$

Lemma 2.2.12 (jsq_eq_L). For all $a \in J$, the square can be expressed in terms of L :

$$a^2 = L_a(a)$$

2.3 IsCommJordan

2.3.1 Multiplication Instance

Definition 2.3.1 (instMul). Define multiplication on J using the Jordan product:

$$a \cdot b := a \circ b$$

Lemma 2.3.2 (mul_eq_jmul). For all $a, b \in J$:

$$a \cdot b = a \circ b$$

2.3.2 Ring Structure

Proposition 2.3.3 (`instNonUnitalNonAssocRing`). J is an instance of `NonUnitalNonAssocRing`, with:

- (Left distributivity) $a \cdot (b + c) = a \cdot b + a \cdot c$
- (Right distributivity) $(a + b) \cdot c = a \cdot c + b \cdot c$
- (Zero multiplication) $0 \cdot a = 0$ and $a \cdot 0 = 0$

Proposition 2.3.4 (`instNonUnitalNonAssocCommRing`). J is an instance of `NonUnitalNonAssocCommRing`, with:

- (Commutativity) $a \cdot b = b \cdot a$

2.3.3 IsCommJordan Instance

Proposition 2.3.5 (`instIsCommJordan`). J is an instance of `IsCommJordan`. The Jordan identity $(a \circ b) \circ a^2 = a \circ (b \circ a^2)$ is exactly what `IsCommJordan` requires.

2.3.4 The Linearized Jordan Identity

Theorem 2.3.6 (`linearized_jordan_mathlib`). For all $a, b, c \in J$:

$$2 \cdot ([L_a, L_{b \cdot c}] + [L_b, L_{c \cdot a}] + [L_c, L_{a \cdot b}]) = 0$$

where $[f, g]$ denotes the Lie bracket (commutator) on `AddMonoid.End`.

Theorem 2.3.7 (`linearized_jordan_jmul`). For all $a, b, c \in J$:

$$2 \cdot ([L_a, L_{b \circ c}] + [L_b, L_{c \circ a}] + [L_c, L_{a \circ b}]) = 0$$

2.3.5 Connection: AddMonoid.End.mulLeft to L Operator

Lemma 2.3.8 (`mulLeft_apply`). For all $a, x \in J$:

$$\text{mulLeft}(a)(x) = a \circ x$$

Lemma 2.3.9 (`mulLeft_eq_L_apply`). For all $a, x \in J$:

$$\text{mulLeft}(a)(x) = L_a(x)$$

Lemma 2.3.10 (`lie_mulLeft_apply`). For all $f, g, x \in J$:

$$[\text{mulLeft}(f), \text{mulLeft}(g)](x) = f \circ (g \circ x) - g \circ (f \circ x)$$

2.4 Quadratic

2.4.1 The Quadratic U Operator

Definition 2.4.1 (`U`). Let J be a Jordan algebra and let $a, x \in J$. The **quadratic U operator** is defined by:

$$U_a(x) := 2 \cdot (a \circ (a \circ x)) - a^2 \circ x$$

Theorem 2.4.2 (`U_def`). For all $a, x \in J$:

$$U_a(x) = 2 \cdot (a \circ (a \circ x)) - a^2 \circ x$$

2.4.2 Basic Properties

Theorem 2.4.3 (U_one). For all $x \in J$:

$$U_1(x) = x$$

Theorem 2.4.4 (U_zero_left). For all $x \in J$:

$$U_0(x) = 0$$

Theorem 2.4.5 (U_zero_right). For all $a \in J$:

$$U_a(0) = 0$$

Theorem 2.4.6 (U_smul). For all $c \in \mathbb{R}$ and $a, x \in J$:

$$U_{c \cdot a}(x) = c^2 \cdot U_a(x)$$

2.4.3 Linearity in the Second Argument

Theorem 2.4.7 (U_add_right). For all $a, x, y \in J$:

$$U_a(x + y) = U_a(x) + U_a(y)$$

Theorem 2.4.8 (U_smul_right). For all $a, x \in J$ and $c \in \mathbb{R}$:

$$U_a(c \cdot x) = c \cdot U_a(x)$$

Definition 2.4.9 (U_linear). Let $a \in J$. Define the $\mathbb{R}$ -linear map $U_a : J \rightarrow_{\mathbb{R}} J$ by:

- toFun(x) := $U_a(x)$
- map_add': U_a is additive (by U_add_right)
- map_smul': U_a respects scalar multiplication (by U_smul_right)

Theorem 2.4.10 (U_linear_apply). For all $a, x \in J$:

$$U_linear(a)(x) = U_a(x)$$

2.4.4 Relationship with L Operator

Theorem 2.4.11 (U_eq_L). For all $a, x \in J$:

$$U_a(x) = 2 \cdot L_a(L_a(x)) - L_{a^2}(x)$$

That is, $U_a = 2L_a^2 - L_{a^2}$.

2.4.5 U Operator Properties

Theorem 2.4.12 (U_self). For all $a \in J$:

$$U_a(a) = a \circ a^2$$

Theorem 2.4.13 (U_idempotent_self). Let $e \in J$ be an idempotent (i.e., $e^2 = e$). Then:

$$U_e(e) = e$$

Theorem 2.4.14 (U_idempotent_comp). Let $e \in J$ be an idempotent. Then U_e is idempotent as an operator:

$$U_linear(e) \circ U_linear(e) = U_linear(e)$$

(Proof requires the fundamental formula.)

Theorem 2.4.15 (U_L_comm). For all $a, x \in J$:

$$U_a(L_a(x)) = L_a(U_a(x))$$

Chapter 3

Operator Identities

3.1 LinearizedJordan

3.1.1 Four-Variable Identity

Theorem 3.1.1 (four_variable_identity). For all $a, b, c, d \in J$:

$$a \circ ((b \circ c) \circ d) + b \circ ((c \circ a) \circ d) + c \circ ((a \circ b) \circ d) = (b \circ c) \circ (a \circ d) + (c \circ a) \circ (b \circ d) + (a \circ b) \circ (c \circ d)$$

Theorem 3.1.2 (operator_formula_apply). For all $a, b, c, d \in J$:

$$a \circ ((b \circ c) \circ d) + b \circ ((c \circ a) \circ d) + c \circ ((a \circ b) \circ d) = (a \circ (b \circ c)) \circ d + b \circ (a \circ (c \circ d)) + c \circ (a \circ (b \circ d))$$

Theorem 3.1.3 (operator_formula). For all $a, b, c \in J$:

$$L_a \circ L_{b \circ c} + L_b \circ L_{c \circ a} + L_c \circ L_{a \circ b} = L_{a \circ (b \circ c)} + L_b \circ L_a \circ L_c + L_c \circ L_a \circ L_b$$

3.1.2 Power Associativity

Theorem 3.1.4 (L_L_jsq_comm). For all $a \in J$, the operators L_a and L_{a^2} commute.

3.1.3 The Fundamental Jordan Identity (Element Form)

Theorem 3.1.5 (fundamental_jordan). For all $a, b \in J$:

$$(a^2 \circ b) \circ a = a^2 \circ (b \circ a)$$

Theorem 3.1.6 (fundamental_jordan'). For all $a, b \in J$:

$$a \circ (a^2 \circ b) = a^2 \circ (a \circ b)$$

Theorem 3.1.7 (fundamental_jordan_original). For all $a, b \in J$:

$$(a \circ b) \circ a^2 = a \circ (b \circ a^2)$$

3.1.4 Power Associativity (Operator Commutation)

Theorem 3.1.8 (L_jpow_comm_L). For all $a \in J$ and $n \in \mathbb{N}$, the operators L_a and L_{a^n} commute.

Theorem 3.1.9 (jpow_add). For all $a \in J$ and $m, n \in \mathbb{N}$:

$$a^m \circ a^n = a^{m+n}$$

Theorem 3.1.10 (jpow_assoc). For all $a \in J$ and $m, n, p \in \mathbb{N}$:

$$(a^m \circ a^n) \circ a^p = a^m \circ (a^n \circ a^p)$$

3.2 FundamentalFormula

Lemma 3.2.1 (linearized_jordan_aux). For all $a, b, c \in J$:

$$\begin{aligned} & (a \circ (b \circ c)) \circ a^2 + (b \circ (a \circ c)) \circ a^2 + (c \circ (a \circ b)) \circ a^2 \\ &= a \circ ((b \circ c) \circ a^2) + b \circ ((a \circ c) \circ a^2) + c \circ ((a \circ b) \circ a^2) \end{aligned}$$

Lemma 3.2.2 (U_U_expand). For all $a, b, x \in J$:

$$U_a(U_b(x)) = 2 \cdot a \circ (a \circ (2 \cdot b \circ (b \circ x) - b^2 \circ x)) - a^2 \circ (2 \cdot b \circ (b \circ x) - b^2 \circ x)$$

3.2.1 The Fundamental Formula

Theorem 3.2.3 (fundamental_formula). (**Fundamental Formula**) For all $a, b \in J$:

$$U_{U_a(b)} = U_a \circ U_b \circ U_a$$

That is, for all $x \in J$:

$$U_{U_a(b)}(x) = U_a(U_b(U_a(x)))$$

3.2.2 Corollaries of the Fundamental Formula

Theorem 3.2.4 (U_jsq). For all $a \in J$:

$$U_{a^2} = U_a^2$$

That is, for all $x \in J$:

$$U_{a^2}(x) = U_a(U_a(x))$$

Theorem 3.2.5 (U_idempotent_comp'). Let $e \in J$ be an idempotent (i.e., $e^2 = e$). Then for all $x \in J$:

$$U_e(U_e(x)) = U_e(x)$$

Theorem 3.2.6 (U_jsq_idempotent). Let $e \in J$ be an idempotent. Then for all $x \in J$:

$$U_{e^2}(x) = U_e(x)$$

3.2.3 Additional Properties from Fundamental Formula

Theorem 3.2.7 (U_jpow_two). For all $a \in J$ and all $x \in J$:

$$U_{a^2}(x) = U_a(U_a(x))$$

Theorem 3.2.8 (fundamental_formula_linear). For all $a, b \in J$, as $\mathbb{R}$ -linear maps:

$$U_{U_a(b)} = U_a \circ U_b \circ U_a$$

3.3 OperatorIdentities

3.3.1 Commutator Bracket

Definition 3.3.1 (opComm). The **commutator** (Lie bracket) of two $\mathbb{R}$ -linear operators $f, g : J \rightarrow J$ is:

$$fg := f \circ g - g \circ f$$

We write f, g for fg .

Lemma 3.3.2 (opComm_apply). For all $\mathbb{R}$ -linear maps $f, g : J \rightarrow J$ and $x \in J$:

$$f, g(x) = f(g(x)) - g(f(x))$$

Lemma 3.3.3 (opComm_skew). For all $\mathbb{R}$ -linear maps $f, g : J \rightarrow J$:

$$f, g = -g, f$$

Lemma 3.3.4 (opComm_self). For all $\mathbb{R}$ -linear maps $f : J \rightarrow J$:

$$f, f = 0$$

3.3.2 Linearized Jordan Identity

Theorem 3.3.5 (linearized_jordan_operator). For all $a, b, c \in J$:

$$2 \cdot (L_a, L_{b \circ c} + L_b, L_{c \circ a} + L_c, L_{a \circ b}) = 0$$

Theorem 3.3.6 (linearized_on_jsq). For all $a, b, c \in J$:

$$\begin{aligned} & 2 \cdot (a \circ ((b \circ c) \circ a^2) - (b \circ c) \circ (a \circ a^2) \\ & + b \circ ((c \circ a) \circ a^2) - (c \circ a) \circ (b \circ a^2) \\ & + c \circ ((a \circ b) \circ a^2) - (a \circ b) \circ (c \circ a^2)) = 0 \end{aligned}$$

Theorem 3.3.7 (linearized_core). For all $a, b, c \in J$:

$$\begin{aligned} & a \circ ((b \circ c) \circ a^2) - (b \circ c) \circ (a \circ a^2) \\ & + b \circ ((c \circ a) \circ a^2) - (c \circ a) \circ (b \circ a^2) \\ & + c \circ ((a \circ b) \circ a^2) - (a \circ b) \circ (c \circ a^2) = 0 \end{aligned}$$

Theorem 3.3.8 (linearized_rearranged). For all $a, b, c \in J$:

$$\begin{aligned} & a \circ ((b \circ c) \circ a^2) + b \circ ((c \circ a) \circ a^2) + c \circ ((a \circ b) \circ a^2) \\ & = (b \circ c) \circ (a \circ a^2) + (c \circ a) \circ (b \circ a^2) + (a \circ b) \circ (c \circ a^2) \end{aligned}$$

3.3.3 Operator Commutator with Squared Element

Theorem 3.3.9 (operator_commutator_jsq). For all $a, b \in J$:

$$L_{a^2}, L_b = 2 \cdot L_a, L_{a \circ b}$$

Theorem 3.3.10 (operator_commutator_jsq_apply). For all $a, b, x \in J$:

$$a^2 \circ (b \circ x) - b \circ (a^2 \circ x) = 2 \cdot (a \circ ((a \circ b) \circ x) - (a \circ b) \circ (a \circ x))$$

3.3.4 Idempotent Operator Identities

Theorem 3.3.11 (opComm_idempotent_eigenspace). Let $e \in J$ be idempotent. Let $a \in J$ with $e \circ a = \lambda \cdot a$ for some $\lambda \in \mathbb{R}$. Then for all $x \in J$:

$$L_e, L_a(x) = e \circ (a \circ x) - a \circ (e \circ x)$$

Theorem 3.3.12 (L_e_L_a_L_e). Let $e \in J$ be idempotent. For all $a \in J$:

$$L_e \circ L_a \circ L_e = L_{e \circ (a \circ e)} + \frac{1}{2} \cdot L_e, L_e, L_a$$

(Currently unproven in the formalization.)

Theorem 3.3.13 (opComm_double_idempotent). Let $e \in J$ be idempotent. For all $a \in J$:

$$L_e, L_e, L_a = 2 \cdot (L_e \circ L_a \circ L_e) - 2 \cdot L_{e \circ (a \circ e)}$$

(Currently unproven in the formalization.)

3.3.5 Useful Commutator Properties

Theorem 3.3.14 (opComm_on_P1). Let $e \in J$ be idempotent. Let $a, b \in J$ with $e \circ a = a$ and $e \circ b = b$ (i.e., $a, b \in P_1(e)$). Then:

$$L_e, L_a(b) = e \circ (a \circ b) - a \circ b$$

Theorem 3.3.15 (opComm_on_P0). Let $e \in J$ be idempotent. Let $a \in J$ with $e \circ a = 0$ (i.e., $a \in P_0(e)$). Then for all $x \in J$:

$$L_e, L_a(x) = e \circ (a \circ x) - a \circ (e \circ x)$$

Chapter 4

Formally Real Jordan Algebras

4.1 Formally Real: Definitions

Definition 4.1.1 (FormallyRealJordan). A Jordan algebra J is **formally real** if for all $n \in \mathbb{N}$ and all functions $a : \text{Fin}(n) \rightarrow J$:

$$\sum_i a_i^2 = 0 \implies \forall i, a_i = 0$$

That is, sums of squares vanish only when all summands vanish.

Theorem 4.1.2 (sq_eq_zero_imp_zero). Let J be a formally real Jordan algebra. If $a \in J$ satisfies $a^2 = 0$, then $a = 0$.

Theorem 4.1.3 (of_sq_eq_zero). Let J be a Jordan algebra. If for all $a \in J$ we have $a^2 = 0 \implies a = 0$, then J is formally real.

Theorem 4.1.4 (formallyReal_iff_sq_eq_zero_imp_zero). A Jordan algebra J is formally real if and only if for all $a \in J$:

$$a^2 = 0 \implies a = 0$$

Definition 4.1.5 (FormallyRealJordan'). A Jordan algebra J satisfies the **FormallyRealJordan'** property if for all $a \in J$:

$$a^2 = 0 \implies a = 0$$

4.1.1 No Nilpotent Elements

Theorem 4.1.6 (jpow_eq_zero_of_ge). Let J be a Jordan algebra and $a \in J$. If $a^n = 0$ and $m \geq n$, then $a^m = 0$.

Theorem 4.1.7 (no_nilpotent_of_formallyReal). Let J be a formally real Jordan algebra. If $a \in J$ and $n \geq 1$ satisfy $a^n = 0$, then $a = 0$.

That is, formally real Jordan algebras have no nilpotent elements.

4.2 Formally Real: Properties

Theorem 4.2.1 (jsq_ne_zero). Let J be a Jordan algebra with the formally real property. For all $a \in J$: if $a \neq 0$, then $a^2 \neq 0$.

Theorem 4.2.2 (jsq_eq_zero_iff). Let J be a Jordan algebra with the formally real property. For all $a \in J$:

$$a^2 = 0 \iff a = 0$$

Theorem 4.2.3 (smul_jone_ne_zero'). Let J be a Jordan algebra. If $\mathbf{1} \neq 0$, then for all $n \in \mathbb{N}$ with $n \neq 0$:

$$n \cdot \mathbf{1} \neq 0$$

This is the Jordan algebra analog of characteristic zero.

Theorem 4.2.4 (jpow_two_eq_zero). Let J be a Jordan algebra with the formally real property. For all $a \in J$: if $a^2 = 0$, then $a = 0$. This is the fundamental “no nilpotent elements” property.

Theorem 4.2.5 (subalgebra_sq_eq_zero). Let J be a Jordan algebra with the formally real property, and let $a \in J$. If $a^2 = 0$, then $a = 0$. Elements of a subalgebra satisfy the formally real property.

4.3 Formally Real: Square Roots

4.3.1 Square Root Predicate

Definition 4.3.1 (IsPositiveSqrt). Let $a, b \in J$. We say b is a **positive square root** of a if

$$b^2 = a \quad \wedge \quad b \in \text{PositiveElement}(J).$$

Definition 4.3.2 (HasPositiveSqrt). An element $a \in J$ has a **positive square root** if

$$\exists b \in J \text{ such that } \text{IsPositiveSqrt}(a, b).$$

4.3.2 Basic Properties

Lemma 4.3.3 (isPositiveSqrt_def). For $a, b \in J$:

$$\text{IsPositiveSqrt}(a, b) \iff b^2 = a \wedge \text{PositiveElement}(b).$$

Theorem 4.3.4 (PositiveElement.of_hasPositiveSqrt). If $a \in J$ has a positive square root, then a is a positive element.

Lemma 4.3.5 (isPositiveSqrt_zero). Zero has zero as its positive square root:

$$\text{IsPositiveSqrt}(0, 0).$$

Lemma 4.3.6 (isPositiveSqrt_jone). If $\mathbf{1}$ is a positive element, then $\mathbf{1}$ is its own positive square root:

$$\text{PositiveElement}(\mathbf{1}) \implies \text{IsPositiveSqrt}(\mathbf{1}, \mathbf{1}).$$

4.3.3 Uniqueness

Theorem 4.3.7 (isPositiveSqrt_unique). Let J be a formally real Jordan algebra. Let $a, b, c \in J$ with

$$\begin{aligned} &\text{IsPositiveSqrt}(a, b), \quad \text{IsPositiveSqrt}(a, c), \\ &b \circ c = c \circ b, \quad b^2 \circ c = c \circ b^2. \end{aligned}$$

Then $b = c$.

4.3.4 Existence

Theorem 4.3.8 (HasPositiveSqrt.of_positiveElement). Let $a \in J$ be a positive element. Then a has a positive square root:

$$\text{PositiveElement}(a) \implies \text{HasPositiveSqrt}(a).$$

4.4 Formally Real: Ordered Cone

4.4.1 Positive Elements

Definition 4.4.1 (PositiveElement). Let J be a Jordan algebra. An element $a \in J$ is **positive** if it is a sum of squares:

$$\text{PositiveElement}(a) := \exists n \in \mathbb{N}, \exists b : \text{Fin}(n) \rightarrow J, \quad a = \sum_{i=0}^{n-1} b_i^2$$

Theorem 4.4.2 (positiveElement_zero). Zero is a positive element:

$$\text{PositiveElement}(0)$$

Theorem 4.4.3 (positiveElement_jsq). For all $a \in J$, the square a^2 is a positive element:

$$\text{PositiveElement}(a^2)$$

Theorem 4.4.4 (positiveElement_add). The sum of positive elements is positive. If $\text{PositiveElement}(a)$ and $\text{PositiveElement}(b)$, then:

$$\text{PositiveElement}(a + b)$$

Theorem 4.4.5 (positiveElement_smul). Positive scalar multiples of positive elements are positive. If $\text{PositiveElement}(a)$ and $r > 0$, then:

$$\text{PositiveElement}(r \cdot a)$$

4.4.2 Positive Cone

Definition 4.4.6 (positiveCone). The **positive cone** in a Jordan algebra J is the convex cone over $\mathbb{R}$ consisting of sums of squares:

$$\text{positiveCone}(J) : \text{ConvexCone } \mathbb{R} J$$

where the carrier is $\{a \in J \mid \text{PositiveElement}(a)\}$, with:

- $\text{smul_mem}'$: for $c > 0$ and x in the cone, $c \cdot x$ is in the cone
- $\text{add_mem}'$: for x, y in the cone, $x + y$ is in the cone

Theorem 4.4.7 (mem_positiveCone). For $a \in J$:

$$a \in \text{positiveCone}(J) \iff \text{PositiveElement}(a)$$

Theorem 4.4.8 (positiveCone_pointed). The positive cone is pointed (contains zero):

$$\text{positiveCone}(J).\text{Pointed}$$

Theorem 4.4.9 (positiveCone_salient). In a formally real Jordan algebra, the positive cone is salient. That is, if J is formally real, then:

$$\text{positiveCone}(J).\text{Salient}$$

In other words, if $a \in \text{positiveCone}(J)$, $a \neq 0$, and $-a \in \text{positiveCone}(J)$, we derive a contradiction.

4.5 Formally Real: Spectrum

4.5.1 Orthogonal Idempotents

Definition 4.5.1 (AreOrthogonal). Two elements $e, f \in J$ are **orthogonal** (in the Jordan sense) if their product is zero:

$$\text{AreOrthogonal}(e, f) := e \circ f = 0$$

Lemma 4.5.2 (AreOrthogonal.symm). For all $e, f \in J$: if $\text{AreOrthogonal}(e, f)$, then $\text{AreOrthogonal}(f, e)$.

Theorem 4.5.3 (jone_sub_idempotent_orthogonal). Let $e \in J$ be an idempotent. Then e and $1 - e$ are orthogonal:

$$\text{IsIdempotent}(e) \implies \text{AreOrthogonal}(e, 1 - e)$$

4.5.2 Complete System of Orthogonal Idempotents

Definition 4.5.4 (PairwiseOrthogonal). A family of elements $(e_i)_{i \in \text{Fin}(n)}$ is **pairwise orthogonal** if:

$$\text{PairwiseOrthogonal}(e) := \forall i, j, i \neq j \implies \text{AreOrthogonal}(e_i, e_j)$$

Definition 4.5.5 (CSOI). A **Complete System of Orthogonal Idempotents (CSOI)** of rank n in a Jordan algebra J consists of:

- $\text{idem} : \text{Fin}(n) \rightarrow J$ — the idempotents
- $\text{is_idem} : \forall i, \text{IsIdempotent}(\text{idem}(i))$ — each element is idempotent
- $\text{orthog} : \text{PairwiseOrthogonal}(\text{idem})$ — pairwise orthogonal
- $\text{complete} : \sum_i \text{idem}(i) = 1$ — sum to identity

4.5.3 Spectral Decomposition

Definition 4.5.6 (SpectralDecomp). A **spectral decomposition** of an element $a \in J$ consists of:

- $n : \mathbb{N}$ — the number of eigenvalues
- $\text{eigenvalues} : \text{Fin}(n) \rightarrow \mathbb{R}$ — the eigenvalues
- $\text{csoi} : \text{CSOI}(J, n)$ — the idempotent system
- $\text{decomp} : a = \sum_i \text{eigenvalues}(i) \cdot \text{csoi}.\text{idem}(i)$ — decomposition equation

Definition 4.5.7 (jordanSpectrum). The **spectrum** of an element $a \in J$ with spectral decomposition sd is:

$$\text{jordanSpectrum}(a, \text{sd}) := \text{range}(\text{sd}.\text{eigenvalues}) \subseteq \mathbb{R}$$

4.5.4 Properties of Orthogonal Idempotents

Theorem 4.5.8 (jmul_orthog_idem). Let $e, f \in J$ be orthogonal elements, and let $r, s \in \mathbb{R}$. Then:

$$(r \cdot e) \circ (s \cdot f) = 0$$

Theorem 4.5.9 (jsq_smul_idem). Let $e \in J$ be an idempotent and $r \in \mathbb{R}$. Then:

$$r \cdot e^2 = r^2 \cdot e$$

4.5.5 Spectral Properties of Squares

Theorem 4.5.10 (spectral_zero_of_eigenvalues_zero). Let $a \in J$ have spectral decomposition sd . If all eigenvalues are zero:

$$(\forall i, \text{sd.eigenvalues}(i) = 0) \implies a = 0$$

Theorem 4.5.11 (spectral_eigenvalues_sq_nonneg). For any spectral decomposition sd of $a \in J$:

$$\forall i, 0 \leq (\text{sd.eigenvalues}(i))^2$$

4.5.6 Formally Real and Spectral Properties

Theorem 4.5.12 (FormallyRealJordan.spectral_sq_eigenvalues_nonneg). Let J be a formally real Jordan algebra and let $a \in J$. If a^2 has a spectral decomposition sd , then all eigenvalues are non-negative:

$$\forall i, 0 \leq \text{sd.eigenvalues}(i)$$

Chapter 5

Peirce Decomposition

5.1 Peirce

5.1.1 Peirce Space Definition

Definition 5.1.1 (PeirceSpace). Let $e \in J$ be an element of a Jordan algebra and $\lambda \in \mathbb{R}$. The Peirce λ -eigenspace for e is:

$$P_\lambda(e) := \{a \in J \mid e \circ a = \lambda \cdot a\}$$

This is a submodule of J over $\mathbb{R}$.

Lemma 5.1.2 (mem_peirceSpace_iff). For all $e, a \in J$ and $\lambda \in \mathbb{R}$:

$$a \in P_\lambda(e) \iff e \circ a = \lambda \cdot a$$

Definition 5.1.3 (PeirceSpace₀, PeirceSpace_{1/2}, PeirceSpace₁). The three standard Peirce spaces are:

- $P_0(e)$ — the 0-eigenspace
- $P_{1/2}(e)$ — the $\frac{1}{2}$ -eigenspace
- $P_1(e)$ — the 1-eigenspace

5.1.2 Basic Peirce Space Properties

Theorem 5.1.4 (peirceSpace_zero_eq_ker). For any $e \in J$:

$$P_0(e) = \ker(L_e)$$

Theorem 5.1.5 (idempotent_in_peirce_one). If e is idempotent (i.e., $e^2 = e$), then $e \in P_1(e)$.

Theorem 5.1.6 (orthogonal_in_peirce_zero). If e and f are orthogonal (i.e., $e \circ f = 0$), then $f \in P_0(e)$.

Theorem 5.1.7 (peirceSpace_disjoint). For any $e \in J$ and distinct eigenvalues $\lambda_1 \neq \lambda_2$, the Peirce spaces are disjoint:

$$P_{\lambda_1}(e) \cap P_{\lambda_2}(e) = \{0\}$$

Theorem 5.1.8 (jone_in_peirce_one). The identity element $\mathbf{1}$ is in its own Peirce 1-space:

$$\mathbf{1} \in P_1(\mathbf{1})$$

Theorem 5.1.9 (peirce_one_left_id). If $a \in P_1(e)$, then e acts as a left identity:

$$e \circ a = a$$

Theorem 5.1.10 (peirce_one_right_id). If $a \in P_1(e)$, then e acts as a right identity:

$$a \circ e = a$$

Theorem 5.1.11 (complement_in_peirce_zero). If e is idempotent, then $1 - e \in P_0(e)$.

5.1.3 Helper Lemmas for Peirce Polynomial

Lemma 5.1.12 (jmul_jmul_e_x_e). For any idempotent e and $x \in J$:

$$(e \circ x) \circ e = e \circ (e \circ x)$$

Lemma 5.1.13 (jmul_nsmul). For any $a, b \in J$ and $n \in \mathbb{N}$:

$$a \circ (n \cdot b) = n \cdot (a \circ b)$$

5.1.4 Peirce Polynomial Identity

Theorem 5.1.14 (peirce_polynomial_identity). For any idempotent e , the left multiplication operator L_e satisfies the Peirce polynomial:

$$L_e \circ \left(L_e - \frac{1}{2} \cdot \text{id} \right) \circ (L_e - \text{id}) = 0$$

Equivalently: $2L_e^3 - 3L_e^2 + L_e = 0$.

This fundamental identity shows that L_e has eigenvalues only in $\{0, \frac{1}{2}, 1\}$.

5.1.5 Peirce Multiplication Rules

Theorem 5.1.15 (peirce_mult_P0_P0). If e is idempotent and $a, b \in P_0(e)$, then:

$$a \circ b \in P_0(e)$$

Theorem 5.1.16 (peirce_mult_P1_P1). If e is idempotent and $a, b \in P_1(e)$, then:

$$a \circ b \in P_1(e)$$

Theorem 5.1.17 (peirce_mult_P0_P1). If e is idempotent, $a \in P_0(e)$, and $b \in P_1(e)$, then:

$$a \circ b = 0$$

Theorem 5.1.18 (peirce_mult_P12_P12). If e is idempotent and $a, b \in P_{1/2}(e)$, then:

$$a \circ b \in P_0(e) \oplus P_1(e)$$

Theorem 5.1.19 (peirce_mult_P0_P12). If e is idempotent, $a \in P_0(e)$, and $b \in P_{1/2}(e)$, then:

$$a \circ b \in P_{1/2}(e)$$

Theorem 5.1.20 (peirce_mult_P1_P12). If e is idempotent, $a \in P_1(e)$, and $b \in P_{1/2}(e)$, then:

$$a \circ b \in P_{1/2}(e)$$

5.1.6 Peirce Projection Operators

Definition 5.1.21 (peirceProj₀). The projection onto $P_0(e)$ is the Lagrange interpolation polynomial for eigenvalue 0:

$$\pi_0 := 2L_e^2 - 3L_e + \text{id} = 2(L_e - \tfrac{1}{2}\text{id})(L_e - \text{id})$$

Definition 5.1.22 (peirceProj_{1/2}). The projection onto $P_{1/2}(e)$ is the Lagrange interpolation polynomial for eigenvalue $\frac{1}{2}$:

$$\pi_{1/2} := -4L_e^2 + 4L_e = -4L_e(L_e - \text{id})$$

Definition 5.1.23 (peirceProj₁). The projection onto $P_1(e)$ is the Lagrange interpolation polynomial for eigenvalue 1:

$$\pi_1 := 2L_e^2 - L_e = 2L_e(L_e - \tfrac{1}{2}\text{id})$$

Theorem 5.1.24 (peirceProj_sum). The three Peirce projections sum to the identity:

$$\pi_0 + \pi_{1/2} + \pi_1 = \text{id}$$

Theorem 5.1.25 (peirceProj_{0_mem}). For any idempotent e and $x \in J$:

$$\pi_0(x) \in P_0(e)$$

Theorem 5.1.26 (peirceProj_{1/2_mem}). For any idempotent e and $x \in J$:

$$\pi_{1/2}(x) \in P_{1/2}(e)$$

Theorem 5.1.27 (peirceProj_{1_mem}). For any idempotent e and $x \in J$:

$$\pi_1(x) \in P_1(e)$$

5.1.7 Peirce Decomposition Theorem

Theorem 5.1.28 (peirce_decomposition). For any idempotent e and $a \in J$, there exist unique elements $a_0, a_{1/2}, a_1 \in J$ such that:

1. $a_0 \in P_0(e)$
2. $a_{1/2} \in P_{1/2}(e)$
3. $a_1 \in P_1(e)$
4. $a = a_0 + a_{1/2} + a_1$

Moreover, $a_\lambda = \pi_\lambda(a)$ for $\lambda \in \{0, \frac{1}{2}, 1\}$.

Theorem 5.1.29 (peirceSpace_iSup_eq_top). For any idempotent e :

$$P_0(e) \oplus P_{1/2}(e) \oplus P_1(e) = J$$

Theorem 5.1.30 (peirce_direct_sum). For any idempotent e , the Peirce spaces form an internal direct sum:

$$J = P_0(e) \oplus P_{1/2}(e) \oplus P_1(e)$$

That is, the decomposition is both complete (spanning) and independent (unique).

5.1.8 PeirceOne Algebraic Structure

Definition 5.1.31 (PeirceOne). For an idempotent e , define $\text{PeirceOne}(e) := P_1(e)$ as a subtype of J .

Proposition 5.1.32 (peirceOneAddCommGroup). $\text{PeirceOne}(e)$ inherits an abelian group structure from the submodule $P_1(e)$.

Proposition 5.1.33 (peirceOneModule). $\text{PeirceOne}(e)$ inherits an $\mathbb{R}$ -module structure from the submodule $P_1(e)$.

Definition 5.1.34 (peirceOneMul). For an idempotent e , define multiplication on $\text{PeirceOne}(e)$ by restricting the Jordan product:

$$a \cdot b := a \circ b$$

This is well-defined by `peirce_mult_P1_P1`.

Definition 5.1.35 (peirceOneOne). For an idempotent e , the identity element of $\text{PeirceOne}(e)$ is e itself.

Lemma 5.1.36 (peirceOne_one_val). For an idempotent e :

$$1_{\text{PeirceOne}(e)} = e$$

Lemma 5.1.37 (peirceOne_mul_val). For an idempotent e and $a, b \in \text{PeirceOne}(e)$:

$$(a \cdot b).\text{val} = a.\text{val} \circ b.\text{val}$$

Theorem 5.1.38 (peirceOne_one_mul). For an idempotent e and $a \in \text{PeirceOne}(e)$:

$$1 \cdot a = a$$

Theorem 5.1.39 (peirceOne_mul_one). For an idempotent e and $a \in \text{PeirceOne}(e)$:

$$a \cdot 1 = a$$

Theorem 5.1.40 (peirceOne_left_distrib). For an idempotent e and $a, b, c \in \text{PeirceOne}(e)$:

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

Theorem 5.1.41 (peirceOne_right_distrib). For an idempotent e and $a, b, c \in \text{PeirceOne}(e)$:

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

Theorem 5.1.42 (peirceOne_mul_comm). For an idempotent e and $a, b \in \text{PeirceOne}(e)$:

$$a \cdot b = b \cdot a$$

Proposition 5.1.43 (peirceOneFiniteDimensional). If J is a finite-dimensional Jordan algebra, then $\text{PeirceOne}(e)$ is finite-dimensional.

5.2 Primitive

5.2.1 Primitive Idempotent Definition

Definition 5.2.1 (IsPrimitive). An idempotent $e \in J$ is **primitive** if:

1. e is idempotent,
2. $e \neq 0$, and
3. for all idempotent f with $e \circ f = f$, either $f = 0$ or $f = e$.

Lemma 5.2.2 (IsPrimitive.isIdempotent). If e is primitive, then e is idempotent.

Lemma 5.2.3 (IsPrimitive.ne_zero). If e is primitive, then $e \neq 0$.

Lemma 5.2.4 (IsPrimitive.sub_eq_zero). Let e be primitive and let f be idempotent with $e \circ f = f$. Then $f = 0$ or $f = e$.

Theorem 5.2.5 (isPrimitive_of_minimal). Let e be idempotent with $e \neq 0$. If for all nonzero idempotent f with $e \circ f = f$ we have $f = e$, then e is primitive.

5.2.2 Properties of Primitive Idempotents

Theorem 5.2.6 (IsPrimitive.orthog_of_ne). Let e be primitive and let f be idempotent with $e \circ f = f$ and $f \neq e$. Then $f = 0$.

Theorem 5.2.7 (IsPrimitive.jmul_complement). Let e be primitive. Then $e \circ (\mathbf{1} - e) = 0$.

5.2.3 Field Classification Results (H-O 2.2.6)

Theorem 5.2.8 (finite_dim_field_over_real). Let F be a finite-dimensional field over $\mathbb{R}$. Then either $F \cong_{\mathbb{R}} \mathbb{R}$ or $F \cong_{\mathbb{R}} \mathbb{C}$.

Theorem 5.2.9 (complex_not_formally_real). $\mathbb{C}$ is not formally real: there exist $n \in \mathbb{N}$ and $a : \text{Fin}(n) \rightarrow \mathbb{C}$ such that $\sum_i a_i^2 = 0$ but $a_i \neq 0$ for some i .

Theorem 5.2.10 (formallyReal_field_is_real). Let F be a finite-dimensional field over $\mathbb{R}$. If F is formally real (i.e., $\sum_i a_i^2 = 0$ implies $a_i = 0$ for all i), then $F \cong_{\mathbb{R}} \mathbb{R}$.

5.2.4 Artinian Ring Structure Theorem (H-O 2.9.3)

Definition 5.2.11 (artinian_reduced_is_product_of_fields). Let R be an Artinian reduced commutative ring. Then

$$R \cong \prod_{I \in \text{MaxSpec}(R)} R/I$$

where each R/I is a field.

Proposition 5.2.12 (artinian_reduced_factor_field). Let R be an Artinian reduced commutative ring and let $I \in \text{MaxSpec}(R)$. Then R/I is a field.

5.2.5 Strongly Connected Idempotents

Definition 5.2.13 (IsStronglyConnected). Two idempotents $e, f \in J$ are **strongly connected** if:

1. e and f are orthogonal, and
2. there exists $v \in J$ such that $e \circ v = \frac{1}{2}v$, $f \circ v = \frac{1}{2}v$, and $v^2 = e + f$.

5.2.6 Structure of Orthogonal Primitives (H-O 2.9.4(iv))

Theorem 5.2.14 (jpow_succ_mem_peirce_one). Let e be idempotent and let $a \in P_1(e)$. Then for all $n \in \mathbb{N}$, $a^{n+1} \in P_1(e)$.

Theorem 5.2.15 (primitive_idempotent_in_P1). Let e be primitive and let f be idempotent with $f \in P_1(e)$. Then $f = 0$ or $f = e$.

Theorem 5.2.16 (peirce_one_sq_nonpos_imp_zero). Let J be formally real, let e be idempotent with $e \neq 0$, and let $x \in P_1(e)$. If $c \leq 0$ and $x^2 = c \cdot e$, then $x = 0$.

Theorem 5.2.17 (lagrange_idempotent_in_peirce_one). Let J be formally real, let e be idempotent, and let $x \in P_1(e)$. Suppose $x^2 = r \cdot x + s \cdot e$ for some $r, s \in \mathbb{R}$ with discriminant $\Delta = r^2 + 4s > 0$. Let $\mu = \frac{r-\sqrt{\Delta}}{2}$ and define

$$f = \frac{1}{\sqrt{\Delta}}(x - \mu \cdot e).$$

Then f is idempotent and $f \in P_1(e)$.

Theorem 5.2.18 (primitive_peirce_one_dim_one). Let J be a finite-dimensional formally real Jordan algebra and let e be primitive. Then

$$\dim_{\mathbb{R}} P_1(e) = 1.$$

Theorem 5.2.19 (primitive_peirce_one_scalar). Let J be a finite-dimensional formally real Jordan algebra and let e be primitive. For every $x \in P_1(e)$, there exists $r \in \mathbb{R}$ such that $x = r \cdot e$.

Theorem 5.2.20 (orthogonal_primitive_peirce_sq). Let J be a finite-dimensional formally real Jordan algebra. Let e, f be primitive and orthogonal, and let $a \in J$ satisfy $e \circ a = \frac{1}{2}a$ and $f \circ a = \frac{1}{2}a$. Then there exists $\mu \geq 0$ such that $a^2 = \mu(e + f)$.

Theorem 5.2.21 (orthogonal_primitive_structure). Let J be a finite-dimensional formally real Jordan algebra. Let e, f be primitive and orthogonal. Then either:

1. for all $a \in J$ with $e \circ a = \frac{1}{2}a$ and $f \circ a = \frac{1}{2}a$, we have $a = 0$, or
2. e and f are strongly connected.

5.2.7 Decomposition into Primitives

Theorem 5.2.22 (exists_primitive_decomp). Let J be a finite-dimensional formally real Jordan algebra. Let e be a nonzero idempotent. Then there exist $k \in \mathbb{N}$ and primitive idempotents $p_1, \dots, p_k$ such that:

1. each p_i is primitive,
2. the p_i are pairwise orthogonal, and
3. $e = \sum_{i=1}^k p_i$.

Theorem 5.2.23 (csoi_refine_primitive). Let J be a finite-dimensional formally real Jordan algebra and let $(e_1, \dots, e_n)$ be a complete system of orthogonal idempotents (CSOI). Then there exists $m \geq n$ and a primitive CSOI $(p_1, \dots, p_m)$ refining it, with each p_i primitive.

5.3 Eigenspace

5.3.1 Eigenspace Definition

Definition 5.3.1 (eigenspace). Let J be a Jordan algebra, $a \in J$, and $\mu \in \mathbb{R}$. The μ -eigenspace of left multiplication by a is defined as:

$$\text{eigenspace}(a, \mu) := \ker(L_a - \mu \cdot \text{id})$$

This is a submodule of J consisting of elements v such that $a \circ v = \mu \cdot v$.

Lemma 5.3.2 (mem_eigenspace_iff). For $a, v \in J$ and $\mu \in \mathbb{R}$:

$$v \in \text{eigenspace}(a, \mu) \iff a \circ v = \mu \cdot v$$

Lemma 5.3.3 (eigenspace_zero). For $a \in J$:

$$\text{eigenspace}(a, 0) = \ker(L_a)$$

Theorem 5.3.4 (eigenspace_eq_peirceSpace). For $a \in J$ and $\mu \in \mathbb{R}$:

$$\text{eigenspace}(a, \mu) = P_\mu(a)$$

5.3.2 Eigenvalue and Spectrum Definitions

Definition 5.3.5 (IsEigenvalue). Let $a \in J$ and $\mu \in \mathbb{R}$. We say μ is an eigenvalue of a if the μ -eigenspace of L_a is nontrivial:

$$\text{IsEigenvalue}(a, \mu) := \text{eigenspace}(a, \mu) \neq \perp$$

Lemma 5.3.6 (isEigenvalue_iff). For $a \in J$ and $\mu \in \mathbb{R}$:

$$\text{IsEigenvalue}(a, \mu) \iff \text{eigenspace}(a, \mu) \neq \perp$$

Lemma 5.3.7 (isEigenvalue_iff_exists_eigenvector). For $a \in J$ and $\mu \in \mathbb{R}$:

$$\text{IsEigenvalue}(a, \mu) \iff \exists v \neq 0, a \circ v = \mu \cdot v$$

Definition 5.3.8 (eigenvalueSet). The eigenvalue set of $a \in J$ is the set of all eigenvalues of L_a :

$$\text{eigenvalueSet}(a) := \{\mu \in \mathbb{R} \mid \text{IsEigenvalue}(a, \mu)\}$$

Lemma 5.3.9 (mem_eigenvalueSet_iff). For $a \in J$ and $\mu \in \mathbb{R}$:

$$\mu \in \text{eigenvalueSet}(a) \iff \text{IsEigenvalue}(a, \mu)$$

5.3.3 Basic Eigenvalue Properties

Theorem 5.3.10 (eigenspace_jone_one). The eigenspace of the identity element $\mathbf{1}$ at eigenvalue 1 is the entire algebra:

$$\text{eigenspace}(\mathbf{1}, 1) = \top$$

Theorem 5.3.11 (isEigenvalue_jone_one). Let J be a nontrivial Jordan algebra. Then 1 is an eigenvalue of $\mathbf{1}$:

$$\text{IsEigenvalue}(\mathbf{1}, 1)$$

Theorem 5.3.12 (eigenvalueSet_jone). Let J be a nontrivial Jordan algebra. The eigenvalue set of $\mathbf{1}$ is exactly $\{1\}$:

$$\text{eigenvalueSet}(\mathbf{1}) = \{1\}$$

5.3.4 Connection to Peirce Decomposition

Theorem 5.3.13 (idempotent_eigenvalues_subset). Let $e \in J$ be an idempotent (i.e., $e^2 = e$). Then the eigenvalues of e are restricted to $\{0, \frac{1}{2}, 1\}$:

$$\text{eigenvalueSet}(e) \subseteq \{0, \frac{1}{2}, 1\}$$

This is a consequence of the Peirce polynomial identity $L_e \circ (L_e - \frac{1}{2}\text{id}) \circ (L_e - \text{id}) = 0$.

5.3.5 Eigenspace Orthogonality

Theorem 5.3.14 (eigenspace_orthogonal). Let J be a Jordan algebra with trace τ . Let $a \in J$ and let $r, s \in \mathbb{R}$ with $r \neq s$. If $v \in \text{eigenspace}(a, r)$ and $w \in \text{eigenspace}(a, s)$, then v and w are orthogonal with respect to the trace inner product:

$$\langle v, w \rangle_\tau = 0$$

Corollary 5.3.15 (eigenspace_traceInner_zero). Let J be a Jordan algebra with trace. For any $a \in J$ and distinct eigenvalues $r \neq s$:

$$\forall v \in \text{eigenspace}(a, r), \forall w \in \text{eigenspace}(a, s), \langle v, w \rangle_\tau = 0$$

5.3.6 Finiteness in Finite Dimensions

Theorem 5.3.16 (eigenvalueSet_finite). Let J be a finite-dimensional Jordan algebra. For any $a \in J$, the eigenvalue set is finite:

$$\text{eigenvalueSet}(a) \text{ is finite}$$

Chapter 6

Structure Theory

6.1 Simple

Definition 6.1.1 (`IsSimpleJordan`). A Jordan algebra J is **simple** if:

- **nontrivial**: There exists $a \in J$ such that $a \neq 0$.
- **ideal_eq_bot_or_top**: For every Jordan ideal I of J , either $I = \perp$ or $I = \top$.

Proposition 6.1.2. If J is a simple Jordan algebra, then J is nontrivial.

Theorem 6.1.3 (`jone_ne_zero`). In a simple Jordan algebra J , the identity element is nonzero: $1 \neq 0$.

Theorem 6.1.4 (`ideal_eq_top_of_ne_bot`). Let J be a simple Jordan algebra and let I be a Jordan ideal of J . If $I \neq \perp$, then $I = \top$.

Theorem 6.1.5 (`ideal_eq_top_of_mem_ne_zero`). Let J be a simple Jordan algebra, let I be a Jordan ideal of J , and let $a \in J$. If $a \in I$ and $a \neq 0$, then $I = \top$.

Theorem 6.1.6 (`ideal_of_one_eq_top`). Let J be a simple Jordan algebra and let I be a Jordan ideal of J . If $1 \in I$, then $I = \top$.

Theorem 6.1.7 (`IsSimpleJordan.mk'`). Let J be a Jordan algebra that is nontrivial. If every Jordan ideal I of J satisfies $I = \perp$ or $I = \top$, then J is simple.

Theorem 6.1.8 (`isSimpleJordan_of_ideal_trichotomy`). Let J be a Jordan algebra. If there exists $a \in J$ with $a \neq 0$, and every Jordan ideal I of J satisfies $I = \perp$ or $I = \top$, then J is simple.

6.2 Semisimple

6.2.1 Ideal Operations

Definition 6.2.1 (`idealSum`). Let I, K be Jordan ideals of J . The **sum of two ideals** is defined as:

$$\text{idealSum}(I, K) := \{x \in J \mid \exists a \in I, \exists b \in K, x = a + b\}$$

This forms a Jordan ideal with:

- **Addition**: If $x = a_1 + b_1$ and $y = a_2 + b_2$ with $a_i \in I, b_i \in K$, then $x + y = (a_1 + a_2) + (b_1 + b_2)$
- **Zero**: $0 = 0 + 0$ with $0 \in I$ and $0 \in K$
- **Scalar multiplication**: $r \cdot x = r \cdot a + r \cdot b$

- Negation: $-x = (-a) + (-b)$
- Jordan multiplication: $y \circ x = (y \circ a) + (y \circ b)$

Lemma 6.2.2 (`mem_idealSum`). For Jordan ideals I, K and $x \in J$:

$$x \in \text{idealSum}(I, K) \iff \exists a \in I, \exists b \in K, x = a + b$$

Lemma 6.2.3 (`le_idealSum_left`). For Jordan ideals I, K : $I \subseteq \text{idealSum}(I, K)$.

Lemma 6.2.4 (`le_idealSum_right`). For Jordan ideals I, K : $K \subseteq \text{idealSum}(I, K)$.

Definition 6.2.5 (`idealInf`). Let I, K be Jordan ideals of J . The **intersection of two ideals** is defined as:

$$\text{idealInf}(I, K) := I \cap K$$

This forms a Jordan ideal.

Lemma 6.2.6 (`mem_idealInf`). For Jordan ideals I, K and $x \in J$:

$$x \in \text{idealInf}(I, K) \iff x \in I \wedge x \in K$$

Lemma 6.2.7 (`idealInf_le_left`). For Jordan ideals I, K : $\text{idealInf}(I, K) \subseteq I$.

Lemma 6.2.8 (`idealInf_le_right`). For Jordan ideals I, K : $\text{idealInf}(I, K) \subseteq K$.

Definition 6.2.9 (`Independent`). Two Jordan ideals I, K are **independent** if their intersection is zero:

$$\text{Independent}(I, K) := \text{idealInf}(I, K) = \perp$$

Lemma 6.2.10 (`independent_iff`). For Jordan ideals I, K :

$$\text{Independent}(I, K) \iff \forall x, (x \in I \rightarrow x \in K \rightarrow x = 0)$$

Definition 6.2.11 (`IsDirectSum`). Two Jordan ideals I, K form a **direct sum decomposition** if they are independent and span J :

$$\text{IsDirectSum}(I, K) := \text{Independent}(I, K) \wedge \text{idealSum}(I, K) = \top$$

Lemma 6.2.12 (`isDirectSum_iff`). For Jordan ideals I, K :

$$\text{IsDirectSum}(I, K) \iff (\forall x, x \in I \rightarrow x \in K \rightarrow x = 0) \wedge (\forall y \in J, \exists a \in I, \exists b \in K, y = a + b)$$

6.2.2 Semisimple Jordan Algebras

Definition 6.2.13 (`IsSemisimpleJordan`). A Jordan algebra J is **semisimple** if:

- (Nontrivial) $\exists a \in J, a \neq 0$
- (Complement property) For every Jordan ideal I of J , there exists a Jordan ideal K such that $\text{IsDirectSum}(I, K)$

Proposition 6.2.14. Every semisimple Jordan algebra J is nontrivial (i.e., $\exists a, b \in J$ with $a \neq b$).

Theorem 6.2.15 (`jone_ne_zero`). In a semisimple Jordan algebra J , the identity is nonzero: $1 \neq 0$.

Proof. Suppose $\mathbf{1} = 0$. By semisimplicity, there exists $a \in J$ with $a \neq 0$. But then:

$$a = a \circ \mathbf{1} = a \circ 0 = 0$$

which contradicts $a \neq 0$. □

Theorem 6.2.16 (IsSimpleJordan.isSemisimpleJordan). Every simple Jordan algebra is semisimple.

Proof. Let J be a simple Jordan algebra and I a Jordan ideal. By simplicity, either $I = \perp$ or $I = \top$.

- If $I = \perp$: Take $K = \top$. Then $\text{idealInf}(\perp, \top) = \perp$ and $\text{idealSum}(\perp, \top) = \top$.
- If $I = \top$: Take $K = \perp$. Then $\text{idealInf}(\top, \perp) = \perp$ and $\text{idealSum}(\top, \perp) = \top$.

In both cases, $\text{IsDirectSum}(I, K)$ holds. □

Theorem 6.2.17 (IsSemisimpleJordan.mk'). Let J be a nontrivial Jordan algebra. If every Jordan ideal I of J has a complement K such that $\text{IsDirectSum}(I, K)$, then J is semisimple.

6.3 Ideal

Definition 6.3.1 (JordanIdeal). A **Jordan ideal** of a Jordan algebra J consists of:

- $\text{carrier} : \mathcal{P}(J)$ — a subset of J
- $\text{add_mem}' : \forall a, b \in \text{carrier}, a + b \in \text{carrier}$
- $\text{zero_mem}' : 0 \in \text{carrier}$
- $\text{smul_mem}' : \forall r \in \mathbb{R}, \forall a \in \text{carrier}, r \cdot a \in \text{carrier}$
- $\text{neg_mem}' : \forall a \in \text{carrier}, -a \in \text{carrier}$
- $\text{jmul_mem}' : \forall a \in J, \forall b \in \text{carrier}, a \circ b \in \text{carrier}$

Proposition 6.3.2. $\text{JordanIdeal } J$ is an instance of SetLike .

Lemma 6.3.3 (mem_carrier). For $I : \text{JordanIdeal } J$ and $x : J$:

$$x \in I.\text{carrier} \iff x \in I$$

Theorem 6.3.4 (add_mem). For $I : \text{JordanIdeal } J$ and $a, b \in J$: if $a \in I$ and $b \in I$, then $a + b \in I$.

Theorem 6.3.5 (zero_mem). For $I : \text{JordanIdeal } J$: $0 \in I$.

Theorem 6.3.6 (smul_mem). For $I : \text{JordanIdeal } J$, $r \in \mathbb{R}$, and $a \in J$: if $a \in I$, then $r \cdot a \in I$.

Theorem 6.3.7 (neg_mem). For $I : \text{JordanIdeal } J$ and $a \in J$: if $a \in I$, then $-a \in I$.

Theorem 6.3.8 (jmul_mem). For $I : \text{JordanIdeal } J$, $a \in J$, and $b \in J$: if $b \in I$, then $a \circ b \in I$.

Theorem 6.3.9 (jmul_mem_left). For $I : \text{JordanIdeal } J$, $a \in J$, and $b \in J$: if $a \in I$, then $a \circ b \in I$.

Theorem 6.3.10 (sub_mem). For $I : \text{JordanIdeal } J$ and $a, b \in J$: if $a \in I$ and $b \in I$, then $a - b \in I$.

Definition 6.3.11 (toAddSubgroup). For $I : \text{JordanIdeal } J$, define $I.\text{toAddSubgroup} : \text{AddSubgroup } J$ with carrier $I.\text{carrier}$, inheriting the additive group structure from I .

Definition 6.3.12 (toSubmodule). For $I : \text{JordanIdeal } J$, define $I.\text{toSubmodule} : \text{Submodule } \mathbb{R} J$ with carrier $I.\text{carrier}$, inheriting the $\mathbb{R}$ -module structure from I .

Proposition 6.3.13. $\text{JordanIdeal } J$ is an instance of Bot .

Definition 6.3.14 (bot). The **zero ideal** $\perp : \text{JordanIdeal } J$ is defined by:

$$\text{carrier} := \{0\}$$

Proposition 6.3.15. $\text{JordanIdeal } J$ is an instance of Top .

Definition 6.3.16 (top). The **whole algebra ideal** $\top : \text{JordanIdeal } J$ is defined by:

$$\text{carrier} := J$$

Lemma 6.3.17 (mem_bot). For $a \in J$: $a \in \perp \iff a = 0$.

Lemma 6.3.18 (mem_top). For $a \in J$: $a \in \top$.

6.4 Subalgebra

Definition 6.4.1 (JordanSubalgebra). A **Jordan subalgebra** of a Jordan algebra J consists of:

- $\text{carrier} : \mathcal{P}(J)$ — a subset of J
- $\text{jone_mem}' : \mathbf{1} \in \text{carrier}$
- $\text{jmul_mem}' : \forall a, b, a \in \text{carrier} \rightarrow b \in \text{carrier} \rightarrow a \circ b \in \text{carrier}$
- $\text{add_mem}' : \forall a, b, a \in \text{carrier} \rightarrow b \in \text{carrier} \rightarrow a + b \in \text{carrier}$
- $\text{smul_mem}' : \forall r \in \mathbb{R}, \forall a, a \in \text{carrier} \rightarrow r \cdot a \in \text{carrier}$
- $\text{neg_mem}' : \forall a, a \in \text{carrier} \rightarrow -a \in \text{carrier}$

Proposition 6.4.2. $\text{JordanSubalgebra } J$ is an instance of SetLike .

Lemma 6.4.3 (mem_carrier). For $S : \text{JordanSubalgebra } J$ and $x : J$:

$$x \in S.\text{carrier} \iff x \in S$$

Theorem 6.4.4 (jone_mem). For $S : \text{JordanSubalgebra } J$:

$$\mathbf{1} \in S$$

Theorem 6.4.5 (jmul_mem). For $S : \text{JordanSubalgebra } J$ and $a, b : J$, if $a \in S$ and $b \in S$, then:

$$a \circ b \in S$$

Theorem 6.4.6 (add_mem). For $S : \text{JordanSubalgebra } J$ and $a, b : J$, if $a \in S$ and $b \in S$, then:

$$a + b \in S$$

Theorem 6.4.7 (smul_mem). For $S : \text{JordanSubalgebra } J$, $r : \mathbb{R}$, and $a : J$, if $a \in S$, then:

$$r \cdot a \in S$$

Theorem 6.4.8 (neg_mem). For $S : \text{JordanSubalgebra } J$ and $a : J$, if $a \in S$, then:

$$-a \in S$$

Theorem 6.4.9 (zero_mem). For $S : \text{JordanSubalgebra } J$:

$$0 \in S$$

Theorem 6.4.10 (sub_mem). For $S : \text{JordanSubalgebra } J$ and $a, b : J$, if $a \in S$ and $b \in S$, then:

$$a - b \in S$$

Definition 6.4.11 (toAddSubgroup). Let $S : \text{JordanSubalgebra } J$. Define $\text{toAddSubgroup}(S) : \text{AddSubgroup } J$ by:

- $\text{carrier} := S.\text{carrier}$
- $\text{add_mem}'$ — inherited from S
- $\text{zero_mem}' := S.\text{zero_mem}$
- $\text{neg_mem}'$ — inherited from S

Definition 6.4.12 (toSubmodule). Let $S : \text{JordanSubalgebra } J$. Define $\text{toSubmodule}(S) : \text{Submodule } \mathbb{R} J$ by:

- $\text{carrier} := S.\text{carrier}$
- $\text{add_mem}'$ — inherited from S
- $\text{zero_mem}' := S.\text{zero_mem}$
- $\text{smul_mem}'$ — inherited from S

6.5 FiniteDimensional

6.5.1 Finite-Dimensional Class

Definition 6.5.1 (FinDimJordanAlgebra). A **finite-dimensional Jordan algebra** is a Jordan algebra J such that J is finite-dimensional as an $\mathbb{R}$ -vector space. Formally, it consists of:

- $\text{finDim} : \text{FiniteDimensional } \mathbb{R} J$

6.5.2 Rank

Definition 6.5.2 (jordanRank). The **Jordan rank** of a finite-dimensional Jordan algebra J is:

$$\text{jordanRank}(J) := \text{finrank}_{\mathbb{R}}(J) : \mathbb{N}$$

Theorem 6.5.3 (jordanRank_pos). Let J be a finite-dimensional Jordan algebra. If J is nontrivial, then:

$$0 < \text{jordanRank}(J)$$

6.5.3 Basis Existence

Theorem 6.5.4 (exists_basis). Let J be a finite-dimensional Jordan algebra. Then there exists $n : \mathbb{N}$ and $b : \text{Fin}(n) \rightarrow J$ such that b is $\mathbb{R}$ -linearly independent and $\text{span}_{\mathbb{R}}(\text{range}(b)) = J$.

Definition 6.5.5 (finBasis). The **canonical basis** of a finite-dimensional Jordan algebra J is:

$$\text{finBasis} : \text{Basis } (\text{Fin}(\text{jordanRank}(J))) \mathbb{R} J$$

6.5.4 Linear Independence of Orthogonal Idempotents

Lemma 6.5.6 (jmul_sum). For any finite set $s : \text{Finset}(\iota)$, element $e \in J$, scalars $r : \iota \rightarrow \mathbb{R}$, and elements $f : \iota \rightarrow J$:

$$e \circ \left(\sum_{i \in s} r_i \cdot f_i \right) = \sum_{i \in s} r_i \cdot (e \circ f_i)$$

Theorem 6.5.7 (linearIndependent_orthog_idem). Let $n : \mathbb{N}$ and $e : \text{Fin}(n) \rightarrow J$. If:

1. $\forall i, e_i$ is idempotent,
2. The family (e_i) is pairwise orthogonal,
3. $\forall i, e_i \neq 0$,

then (e_i) is $\mathbb{R}$ -linearly independent.

6.5.5 CSOI Cardinality Bound

Theorem 6.5.8 (csoi_card_le_rank_of_nonzero). Let J be a finite-dimensional Jordan algebra and let c be a CSOI of cardinality n . If $\forall i, c.\text{idem}(i) \neq 0$, then:

$$n \leq \text{jordanRank}(J)$$

Theorem 6.5.9 (jone_ne_zero). Let J be a nontrivial Jordan algebra. Then:

$$\mathbf{1} \neq 0$$

Theorem 6.5.10 (csoi_exists_nonzero). Let J be a nontrivial Jordan algebra and let c be a CSOI of cardinality n with $0 < n$. Then:

$$\exists i, c.\text{idem}(i) \neq 0$$

6.6 Trace

6.6.1 Trace Class

Definition 6.6.1 (JordanTrace). A **Jordan trace** on a Jordan algebra J is a structure consisting of:

- $\text{trace} : J \rightarrow \mathbb{R}$ — the trace functional
- $\text{trace_add} : \forall a, b \in J, (a + b) = (a) + (b)$
- $\text{trace_smul} : \forall r \in \mathbb{R}, \forall a \in J, (r \cdot a) = r \cdot (a)$
- $\text{trace_jmul_comm} : \forall a, b \in J, (a \circ b) = (b \circ a)$
- $\text{trace_L_selfadjoint} : \forall a, v, w \in J, ((a \circ v) \circ w) = (v \circ (a \circ w))$
- $\text{trace_jone_pos} : 0 < (\mathbf{1})$

6.6.2 Basic Trace Properties

Theorem 6.6.2 (trace_zero). $(0) = 0$.

Theorem 6.6.3 (trace_neg). For all $a \in J$:

$$(-a) = -(a)$$

Theorem 6.6.4 (trace_sub). For all $a, b \in J$:

$$(a - b) = (a) - (b)$$

6.6.3 Trace Inner Product

Definition 6.6.5 (traceInner). The **trace inner product** is defined by:

$$\tau(a, b) := (a \circ b)$$

Theorem 6.6.6 (traceInner_symm). For all $a, b \in J$:

$$\tau(a, b) = \tau(b, a)$$

Theorem 6.6.7 (traceInner_add_left). For all $a, b, c \in J$:

$$\tau(a + b, c) = \tau(a, c) + \tau(b, c)$$

Theorem 6.6.8 (traceInner_add_right). For all $a, b, c \in J$:

$$\tau(a, b + c) = \tau(a, b) + \tau(a, c)$$

Theorem 6.6.9 (traceInner_smul_left). For all $r \in \mathbb{R}$ and $a, b \in J$:

$$\tau(r \cdot a, b) = r \cdot \tau(a, b)$$

Theorem 6.6.10 (traceInner_smul_right). For all $r \in \mathbb{R}$ and $a, b \in J$:

$$\tau(a, r \cdot b) = r \cdot \tau(a, b)$$

Theorem 6.6.11 (traceInner_zero_left). For all $a \in J$:

$$\tau(0, a) = 0$$

Theorem 6.6.12 (traceInner_zero_right). For all $a \in J$:

$$\tau(a, 0) = 0$$

Theorem 6.6.13 (traceInner_neg_left). For all $a, b \in J$:

$$\tau(-a, b) = -\tau(a, b)$$

Theorem 6.6.14 (traceInner_neg_right). For all $a, b \in J$:

$$\tau(a, -b) = -\tau(a, b)$$

Theorem 6.6.15 (traceInner_jmul_left). For all $a, v, w \in J$:

$$\tau(a \circ v, w) = \tau(v, a \circ w)$$

That is, L_a is self-adjoint with respect to the trace inner product.

6.6.4 Non-degeneracy

Definition 6.6.16 (FormallyRealTrace). A **formally real trace** on a Jordan algebra J extends JordanTrace with:

- trace_jsq_nonneg: $\forall a \in J, 0 \leq (a^2)$
- trace_jsq_eq_zero_iff: $\forall a \in J, (a^2) = 0 \iff a = 0$

Theorem 6.6.17 (traceInner_self_nonneg). For all $a \in J$:

$$0 \leq \tau(a, a)$$

Theorem 6.6.18 (traceInner_self_eq_zero_iff). For all $a \in J$:

$$\tau(a, a) = 0 \iff a = 0$$

Theorem 6.6.19 (traceInner_self_pos). If $a \neq 0$, then:

$$0 < \tau(a, a)$$

Theorem 6.6.20 (traceInner_nondegenerate). If $\tau(a, b) = 0$ for all $b \in J$, then $a = 0$.

Chapter 7

Spectral Theory

7.1 SpectralTheorem

7.1.1 Spectrum Definition

Definition 7.1.1 (spectrum). The **spectrum** of an element $a \in J$ is the set of eigenvalues of L_a :

$$\text{spectrum}(a) := \text{eigenvalueSet}(a)$$

In finite dimensions, this is a finite set of real numbers.

Lemma 7.1.2 (mem_spectrum_iff). For $a \in J$ and $r \in \mathbb{R}$:

$$r \in \text{spectrum}(a) \iff \text{IsEigenvalue}(a, r)$$

Theorem 7.1.3 (spectrum_finite). Let J be a finite-dimensional Jordan algebra. For any $a \in J$, $\text{spectrum}(a)$ is finite.

7.1.2 Spectral Decomposition Existence

Theorem 7.1.4 (spectral_decomposition_exists). Let J be a finite-dimensional formally real Jordan algebra with trace. For every $a \in J$, there exists a spectral decomposition sd of a such that for all i , $\text{sd.csoi.idem}(i)$ is primitive.

Theorem 7.1.5 (spectral_decomposition_finset). Let J be a finite-dimensional formally real Jordan algebra with trace. For every $a \in J$, there exist:

- A finite set $S \subseteq \mathbb{R}$
- A function $e : \mathbb{R} \rightarrow J$

such that:

1. For all $r \in S$, $e(r)$ is idempotent
2. For all $r, s \in S$ with $r \neq s$, $e(r)$ and $e(s)$ are orthogonal
3. $\sum_{r \in S} e(r) = \mathbf{1}$
4. $a = \sum_{r \in S} r \cdot e(r)$

7.1.3 Uniqueness Results

Theorem 7.1.6 (spectrum_eq_eigenvalueSet). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$ and any spectral decomposition $\mathbf{sd}$ of a :

$$\text{jordanSpectrum}(a, \mathbf{sd}) = \text{spectrum}(a)$$

Theorem 7.1.7 (spectral_decomp_eigenvalues_eq_spectrum). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$ and spectral decomposition $\mathbf{sd}$:

$$\text{range}(\mathbf{sd}.\text{eigenvalues}) = \text{spectrum}(a)$$

7.1.4 Square Eigenvalue Theorem

Lemma 7.1.8 (sum_jmul). For a finite index set ι , elements $f : \iota \rightarrow J$, scalars $r : \iota \rightarrow \mathbb{R}$, and $e \in J$:

$$\left(\sum_{i \in s} r_i \cdot f_i \right) \circ e = \sum_{i \in s} r_i \cdot (f_i \circ e)$$

Theorem 7.1.9 (jsq_sum_orthog_idem). Let c be a complete system of orthogonal idempotents of size n , and let $\text{coef} : \text{Fin}(n) \rightarrow \mathbb{R}$. Then:

$$\left(\sum_i \text{coef}_i \cdot c.\text{idem}(i) \right)^2 = \sum_i \text{coef}_i^2 \cdot c.\text{idem}(i)$$

Theorem 7.1.10 (spectral_sq). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$ with spectral decomposition $\mathbf{sd}$, there exists a spectral decomposition $\mathbf{sd_sq}$ of a^2 with the same size:

$$\mathbf{sd_sq}.n = \mathbf{sd}.n$$

The spectral decomposition of a^2 uses the same complete system of orthogonal idempotents with squared eigenvalues. If $a = \sum_i \lambda_i \cdot e_i$ with orthogonal idempotents, then:

$$a^2 = \left(\sum_i \lambda_i \cdot e_i \right)^2 = \sum_i \lambda_i^2 \cdot e_i$$

since $e_i \circ e_j = 0$ for $i \neq j$ and $e_i^2 = e_i$.

Theorem 7.1.11 (spectrum_sq). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$:

$$\text{spectrum}(a^2) = \{r^2 : r \in \text{spectrum}(a)\}$$

7.1.5 Positivity of Square Eigenvalues

Theorem 7.1.12 (sq_eigenvalues_nonneg). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$ and any spectral decomposition $\mathbf{sd}$ of a^2 :

$$\forall i, 0 \leq \mathbf{sd}.\text{eigenvalues}(i)$$

Theorem 7.1.13 (spectrum_sq_nonneg). Let J be a finite-dimensional formally real Jordan algebra with trace. For $a \in J$:

$$\forall r \in \text{spectrum}(a^2), 0 \leq r$$

Chapter 8

Reversibility

8.1 Reversible

Definition 8.1.1 (IsReversible). A Jordan algebra J is **reversible** if there exists:

- An associative algebra A over $\mathbb{R}$
- An $\mathbb{R}$ -linear map $f : J \rightarrow A$

such that:

1. (Jordan product preservation) For all $a, b \in J$:

$$f(a \circ b) = \frac{1}{2}(f(a) \cdot f(b) + f(b) \cdot f(a))$$

2. (Identity preservation) $f(\mathbf{1}) = 1_A$

3. (Injectivity) f is injective

4. (Closure under reversal) For all $a, b, c \in J$, there exists $d \in J$ such that:

$$f(d) = f(a) \cdot f(b) \cdot f(c) + f(c) \cdot f(b) \cdot f(a)$$

Chapter 9

Classification

9.1 Classification: Simple Types

Definition 9.1.1 (SimpleType). The **simple types** of finite-dimensional simple formally real Jordan algebras are given by the inductive type:

- $\text{typel}(n) : \text{SimpleType} \text{ — } H_n(\mathbb{R})$: real symmetric matrices
- $\text{typell}(n) : \text{SimpleType} \text{ — } H_n(\mathbb{C})$: complex Hermitian matrices
- $\text{typelll}(n) : \text{SimpleType} \text{ — } H_n(\mathbb{H})$: quaternionic Hermitian matrices
- $\text{typelV}(n) : \text{SimpleType} \text{ — } V_n$: spin factors ($n \geq 2$)
- $\text{typeV} : \text{SimpleType} \text{ — } H_3(\mathbb{O})$: Albert algebra (exceptional)

This type has decidable equality.

9.1.1 Properties of Simple Types

Definition 9.1.2 (dimension). The **dimension** of a simple type is defined by:

$$\text{dimension}(t) := \begin{cases} \frac{n(n+1)}{2} & \text{if } t = \text{typel}(n) \\ n^2 & \text{if } t = \text{typell}(n) \\ n(2n-1) & \text{if } t = \text{typelll}(n) \\ n & \text{if } t = \text{typelV}(n) \\ 27 & \text{if } t = \text{typeV} \end{cases}$$

Definition 9.1.3 (isReversible). A simple type is **reversible** (embeds into an associative algebra with reversal) if and only if it is not of type V:

$$\text{isReversible}(t) := \begin{cases} \text{false} & \text{if } t = \text{typeV} \\ \text{true} & \text{otherwise} \end{cases}$$

Definition 9.1.4 (isExceptional). A simple type is **exceptional** (not special) if and only if it is of type V:

$$\text{isExceptional}(t) := \begin{cases} \text{true} & \text{if } t = \text{typeV} \\ \text{false} & \text{otherwise} \end{cases}$$

Definition 9.1.5 (minIndex). The **minimum index** for which each type is defined:

$$\text{minIndex}(t) := \begin{cases} 1 & \text{if } t = \text{typeI}(n) \\ 1 & \text{if } t = \text{typeII}(n) \\ 1 & \text{if } t = \text{typeIII}(n) \\ 2 & \text{if } t = \text{typeIV}(n) \\ 0 & \text{if } t = \text{typeV} \end{cases}$$

Note: $V_1 \cong \mathbb{R}$ is not considered a “spin” factor, hence the minimum index of 2 for type IV.

9.2 Classification: Real Symmetric

Theorem 9.2.1 (nontrivial). Let n be a nonempty finite type. Then the Jordan algebra of real symmetric matrices $H_n(\mathbb{R})$ is nontrivial:

$$\exists A \in H_n(\mathbb{R}), \quad A \neq 0$$

Proposition 9.2.2 (isSimple). Let n be a nonempty finite type with decidable equality. Then the Jordan algebra of real symmetric matrices $H_n(\mathbb{R})$ is a simple Jordan algebra:

$$\text{IsSimpleJordan}(H_n(\mathbb{R}))$$

9.3 Classification: Complex Hermitian

9.3.1 Nontriviality

Theorem 9.3.1 (nontrivial). Let n be a nonempty finite type. Then there exists a complex Hermitian matrix $A \in H_n(\mathbb{C})$ such that $A \neq 0$.

9.3.2 Simplicity

Proposition 9.3.2 (isSimple). Let n be a nonempty finite type with decidable equality. Then the Jordan algebra of complex Hermitian matrices $H_n(\mathbb{C})$ is a simple Jordan algebra.

Chapter 10

Matrix Jordan Algebras

10.1 Matrix: Jordan Product

10.1.1 Basic Jordan Product

Let n be a finite type with decidable equality, and let R be a field of characteristic zero.

Definition 10.1.1 (jordanMul). The **Jordan product** on matrices $A, B \in M_n(R)$ is defined by:

$$A \circ B := \frac{1}{2}(AB + BA)$$

Lemma 10.1.2 (jordanMul_def). For all $A, B \in M_n(R)$:

$$A \circ B = \frac{1}{2}(AB + BA)$$

Theorem 10.1.3 (jordanMul_comm). The Jordan product is commutative. For all $A, B \in M_n(R)$:

$$A \circ B = B \circ A$$

Lemma 10.1.4 (jordanMul_one). The identity matrix is the identity for the Jordan product. For all $A \in M_n(R)$:

$$A \circ I = A$$

Lemma 10.1.5 (one_jordanMul). For all $A \in M_n(R)$:

$$I \circ A = A$$

Lemma 10.1.6 (jordanMul_zero). For all $A \in M_n(R)$:

$$A \circ 0 = 0$$

Lemma 10.1.7 (zero_jordanMul). For all $A \in M_n(R)$:

$$0 \circ A = 0$$

Theorem 10.1.8 (jordanMul_self). The Jordan square equals the matrix square. For all $A \in M_n(R)$:

$$A \circ A = A^2$$

10.1.2 Hermitian Matrices

Let R be a field of characteristic zero with a star ring structure.

Theorem 10.1.9 (IsHermitian.jordanMul). Hermitian matrices are closed under the Jordan product. If $A, B \in M_n(R)$ are Hermitian (i.e., $A^* = A$ and $B^* = B$), then $A \circ B$ is also Hermitian:

$$(A \circ B)^* = A \circ B$$

10.2 Matrix: Instance

Definition 10.2.1 (HermitianMatrix). Let n be a finite type and R be an additive group with star structure. Define:

$$\text{HermitianMatrix}(n, R) := \text{selfAdjoint}(\text{Matrix}(n, n, R))$$

i.e., the type of self-adjoint (Hermitian) $n \times n$ matrices over R .

10.2.1 Jordan Product on Hermitian Matrices

Definition 10.2.2 (jmul). Let $A, B : \text{HermitianMatrix}(n, R)$. Define the **Jordan product**:

$$A \circ B := \frac{1}{2}(AB + BA)$$

which is again Hermitian.

Lemma 10.2.3 (jmul_val). For $A, B : \text{HermitianMatrix}(n, R)$:

$$(A \circ B).\text{val} = \text{jordanMul}(A.\text{val}, B.\text{val})$$

Theorem 10.2.4 (jmul_comm). The Jordan product on Hermitian matrices is commutative:

$$A \circ B = B \circ A$$

Definition 10.2.5 (one). The **identity element** in $\text{HermitianMatrix}(n, R)$ is:

$$\mathbf{1} := I_n$$

where I_n is the identity matrix, which is Hermitian.

Lemma 10.2.6 (one_val).

$$\mathbf{1}.\text{val} = 1$$

Theorem 10.2.7 (jmul_one). For all $A : \text{HermitianMatrix}(n, R)$:

$$A \circ \mathbf{1} = A$$

Theorem 10.2.8 (one_jmul). For all $A : \text{HermitianMatrix}(n, R)$:

$$\mathbf{1} \circ A = A$$

Theorem 10.2.9 (jmul_add). The Jordan product distributes over addition:

$$A \circ (B + C) = (A \circ B) + (A \circ C)$$

Theorem 10.2.10 (jmul_smul). The Jordan product respects scalar multiplication: for $r \in \mathbb{R}$,

$$(r \cdot A) \circ B = r \cdot (A \circ B)$$

10.2.2 Jordan Identity

Theorem 10.2.11 (jordanMul_jordan_identity). For matrices $A, B : \text{Matrix}(n, n, R)$, the Jordan identity holds:

$$(A \circ B) \circ (A \circ A) = A \circ (B \circ (A \circ A))$$

where $A \circ B = \frac{1}{2}(AB + BA)$. Both sides expand to:

$$\frac{1}{4}(ABA^2 + BA^3 + A^3B + A^2BA)$$

Theorem 10.2.12 (jordan_identity). For Hermitian matrices $A, B : \text{HermitianMatrix}(n, R)$:

$$(A \circ B) \circ A^2 = A \circ (B \circ A^2)$$

10.2.3 Jordan Algebra Instance

Proposition 10.2.13 (jordanAlgebraHermitianMatrix). Let n be a finite decidable type and R a field with star ring structure, characteristic zero, and $\mathbb{R}$ -algebra structure with star module. Then $\text{HermitianMatrix}(n, R)$ is a **Jordan algebra** with:

- Product: $A \circ B = \frac{1}{2}(AB + BA)$
- Commutativity: $A \circ B = B \circ A$
- Jordan identity: $(A \circ B) \circ A^2 = A \circ (B \circ A^2)$
- Identity: $\mathbf{1} = I_n$
- Left identity: $\mathbf{1} \circ A = A$
- Distributivity: $A \circ (B + C) = (A \circ B) + (A \circ C)$
- Scalar compatibility: $(r \cdot A) \circ B = r \cdot (A \circ B)$

10.3 Matrix: Formally Real

This section establishes that Hermitian matrices over an RCLike field form a formally real Jordan algebra.

10.3.1 Jordan Square as Matrix Square

Theorem 10.3.1 (jsq_val). For a Hermitian matrix $A \in \text{HermitianMatrix}(n, \mathbb{K})$, the Jordan square equals the matrix square:

$$(A^2).\text{val} = A.\text{val} \cdot A.\text{val}$$

Theorem 10.3.2 (sq_eq_conjTranspose_mul). For a Hermitian matrix A , the matrix square equals the conjugate-transpose product:

$$A.\text{val} \cdot A.\text{val} = A.\text{val}^H \cdot A.\text{val}$$

Theorem 10.3.3 (sq_posSemidef). For a Hermitian matrix A , the matrix square is positive semidefinite:

$$(A.\text{val} \cdot A.\text{val}).\text{PosSemidef}$$

Theorem 10.3.4 (jsq_nonneg). The Jordan square of a Hermitian matrix is nonnegative in the matrix order:

$$0 \leq (A^2).\text{val}$$

10.3.2 Zero Characterization

Theorem 10.3.5 (sq_eq_zero_imp_zero). For Hermitian matrices, if the square vanishes then the matrix is zero:

$$A.\text{val} \cdot A.\text{val} = 0 \implies A = 0$$

Theorem 10.3.6 (jsq_eq_zero_imp_zero). For Hermitian matrices, if the Jordan square vanishes then the matrix is zero:

$$A^2 = 0 \implies A = 0$$

10.3.3 Sum of Squares Characterization

Theorem 10.3.7 (sum_jsq_val). For a family $a : \text{Fin}(m) \rightarrow \text{HermitianMatrix}(n, \mathbb{K})$, the sum of Jordan squares corresponds to the sum of matrix squares:

$$\left(\sum_i a_i^2 \right).\text{val} = \sum_i a_i.\text{val} \cdot a_i.\text{val}$$

Lemma 10.3.8 (each_sq_nonneg). Each matrix square in a family is nonnegative:

$$\forall i, 0 \leq a_i.\text{val} \cdot a_i.\text{val}$$

Theorem 10.3.9 (sum_sq_eq_zero_imp_all_zero). If a sum of squares of Hermitian matrices is zero, each term is zero. This is the key property making Hermitian matrices formally real:

$$\sum_i a_i.\text{val} \cdot a_i.\text{val} = 0 \implies \forall i, a_i = 0$$

10.3.4 FormallyRealJordan Instance

Proposition 10.3.10 (formallyRealJordanHermitianMatrix). Hermitian matrices over an RCLike field form a formally real Jordan algebra:

$$\text{HermitianMatrix}(n, \mathbb{K}) \text{ is an instance of FormallyRealJordan}$$

The instance satisfies the formally real property: for any finite family $a : \text{Fin}(m) \rightarrow \text{HermitianMatrix}(n, \mathbb{K})$,

$$\sum_i a_i^2 = 0 \implies \forall i, a_i = 0$$

10.4 Matrix: Trace

10.4.1 Trace Inner Product Definition

Definition 10.4.1 (traceInner). The **trace inner product** on Hermitian matrices is defined as follows. Let $A, B : \text{HermitianMatrix}(n, \mathbb{K})$. Define:

$$\text{traceInner}(A, B) := (A \cdot B)$$

Lemma 10.4.2 (traceInner_def). For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{traceInner}(A, B) = (A \cdot B)$$

10.4.2 Basic Properties

Theorem 10.4.3 (`traceInner_comm`). The trace inner product is commutative. For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, B) = \text{tracelInner}(B, A)$$

Theorem 10.4.4 (`traceInner_add_left`). The trace inner product is additive in the first argument. For all $A, B, C : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A + B, C) = \text{tracelInner}(A, C) + \text{tracelInner}(B, C)$$

Theorem 10.4.5 (`traceInner_add_right`). The trace inner product is additive in the second argument. For all $A, B, C : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, B + C) = \text{tracelInner}(A, B) + \text{tracelInner}(A, C)$$

Theorem 10.4.6 (`traceInner_zero_left`). For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(0, A) = 0$$

Theorem 10.4.7 (`traceInner_zero_right`). For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, 0) = 0$$

10.4.3 Self Inner Product

Theorem 10.4.8 (`traceInner_self`). For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, A) = (A^2)$$

Theorem 10.4.9 (`sq_eq_conjTranspose_mul'`). For Hermitian A , we have $A^2 = A^\dagger A$. That is, for all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$A \cdot A = A^\dagger \cdot A$$

Theorem 10.4.10 (`traceInner_self_conj`). The self inner product is real (equals its conjugate). For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\overline{\text{tracelInner}(A, A)} = \text{tracelInner}(A, A)$$

Theorem 10.4.11 (`traceInner_self_nonneg`). The self inner product is nonnegative. For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$0 \leq \text{tracelInner}(A, A)$$

Theorem 10.4.12 (`traceInner_self_eq_zero`). The self inner product is zero if and only if the matrix is zero. For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, A) = 0 \iff A = 0$$

10.4.4 Trace of Jordan Product

Theorem 10.4.13 (`traceInner_eq_trace_jmul`). The trace inner product equals the trace of the Jordan product. For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInner}(A, B) = (A \circ B)$$

where the Jordan product is $A \circ B = \frac{1}{2}(AB + BA)$.

Theorem 10.4.14 (`traceInner_conj`). The trace inner product is real-valued for all Hermitian pairs. For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\overline{\text{tracelInner}(A, B)} = \text{tracelInner}(A, B)$$

10.4.5 Real Inner Product

Definition 10.4.15 (traceInnerReal). The **real-valued trace inner product** is defined as follows. Let $A, B : \text{HermitianMatrix}(n, \mathbb{K})$. Define:

$$\text{tracelInnerReal}(A, B) := \Re(\text{tracelInner}(A, B))$$

where $\Re$ denotes the real part.

Lemma 10.4.16 (traceInnerReal_def). For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInnerReal}(A, B) = \Re(\text{tracelInner}(A, B))$$

Theorem 10.4.17 (traceInnerReal_comm). The real trace inner product is commutative. For all $A, B : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInnerReal}(A, B) = \text{tracelInnerReal}(B, A)$$

Theorem 10.4.18 (traceInnerReal_add_left). The real trace inner product is additive. For all $A, B, C : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInnerReal}(A + B, C) = \text{tracelInnerReal}(A, C) + \text{tracelInnerReal}(B, C)$$

Theorem 10.4.19 (traceInnerReal_self_nonneg). The real trace inner product self is nonnegative. For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$0 \leq \text{tracelInnerReal}(A, A)$$

Theorem 10.4.20 (traceInnerReal_self_eq_zero). The real trace inner product self equals zero if and only if $A = 0$. For all $A : \text{HermitianMatrix}(n, \mathbb{K})$:

$$\text{tracelInnerReal}(A, A) = 0 \iff A = 0$$

10.5 Matrix: Real Hermitian

This section establishes the equivalence between Hermitian and symmetric matrices over $\mathbb{R}$, and defines the type of real symmetric matrices as a Jordan algebra.

10.5.1 Hermitian and Symmetric Equivalence for Real Matrices

Theorem 10.5.1 (isHermitian_iff_isSymm). Let $A : \text{Matrix}(n, n, \mathbb{R})$. Then A is Hermitian if and only if A is symmetric:

$$A^* = A \iff A^\top = A$$

Theorem 10.5.2 (conjTranspose_eq_transpose_real). For any real matrix $A : \text{Matrix}(m, n, \mathbb{R})$:

$$A^* = A^\top$$

Lemma 10.5.3 (IsSymm.isHermitian). If $A : \text{Matrix}(n, n, \mathbb{R})$ is symmetric, then A is Hermitian.

Lemma 10.5.4 (IsHermitian.isSymm_of_real). If $A : \text{Matrix}(n, n, \mathbb{R})$ is Hermitian, then A is symmetric.

10.5.2 Real Symmetric Matrix Type

Definition 10.5.5 (RealSymmetricMatrix). Define $\text{RealSymmetricMatrix}(n)$ as an abbreviation for $\text{HermitianMatrix}(n, \mathbb{R})$.

Over $\mathbb{R}$, this coincides with the set of matrices satisfying $A^\top = A$.

10.5.3 Basic Properties

Theorem 10.5.6 (transpose_eq). For any $A : \text{RealSymmetricMatrix}(n)$:

$$A^\top = A$$

Definition 10.5.7 (ofSymm). Let $A : \text{Matrix}(n, n, \mathbb{R})$ and let h be a proof that A is symmetric. Define:

$$\text{ofSymm}(A, h) : \text{RealSymmetricMatrix}(n)$$

as the real symmetric matrix with underlying matrix A .

Lemma 10.5.8 (ofSymm_val). For any symmetric matrix $A : \text{Matrix}(n, n, \mathbb{R})$ with proof h :

$$\text{ofSymm}(A, h).\text{val} = A$$

Theorem 10.5.9 (isSymm). For any $A : \text{RealSymmetricMatrix}(n)$, the underlying matrix $A.\text{val}$ is symmetric.

10.5.4 Instances

Proposition 10.5.10. $\text{RealSymmetricMatrix}(n)$ is an instance of JordanAlgebra .

Proposition 10.5.11. $\text{RealSymmetricMatrix}(n)$ is an instance of $\text{FormallyRealJordan}$.

10.5.5 Special Matrices

Definition 10.5.12 (one). The identity matrix in $\text{RealSymmetricMatrix}(n)$:

$$\text{one} := \text{HermitianMatrix.one}$$

Definition 10.5.13 (zero). The zero matrix in $\text{RealSymmetricMatrix}(n)$:

$$\text{zero} := 0$$

Definition 10.5.14 (diagonal). Let $d : n \rightarrow \mathbb{R}$ be a function. Define the diagonal matrix:

$$\text{diagonal}(d) : \text{RealSymmetricMatrix}(n)$$

as the real symmetric matrix with diagonal entries given by d .

Lemma 10.5.15 (diagonal_val). For any $d : n \rightarrow \mathbb{R}$:

$$\text{diagonal}(d).\text{val} = \text{Matrix.diagonal}(d)$$

10.5.6 Trace Inner Product

Definition 10.5.16 (traceInner). For $A, B : \text{RealSymmetricMatrix}(n)$, define the trace inner product:

$$\langle A, B \rangle_{\text{tr}} := \text{HermitianMatrix.traceInnerReal}(A, B)$$

Theorem 10.5.17 (traceInner_comm). For all $A, B : \text{RealSymmetricMatrix}(n)$:

$$\langle A, B \rangle_{\text{tr}} = \langle B, A \rangle_{\text{tr}}$$

Theorem 10.5.18 (traceInner_self_nonneg). For all $A : \text{RealSymmetricMatrix}(n)$:

$$0 \leq \langle A, A \rangle_{\text{tr}}$$

10.6 Matrix: Complex Hermitian

Definition 10.6.1 (ComplexHermitianMatrix). Let n be a finite type. The **complex Hermitian matrices** are defined as:

$$\text{ComplexHermitianMatrix}(n) := \text{HermitianMatrix}(n, \mathbb{C})$$

These are matrices A satisfying $A^\dagger = A$ where $A^\dagger$ denotes the conjugate transpose.

10.6.1 Basic Properties

Theorem 10.6.2 (conjTranspose_eq). Let $A : \text{ComplexHermitianMatrix}(n)$. Then:

$$A^\dagger = A$$

Definition 10.6.3 (ofHermitian). Let $A : \text{Matrix}(n, n, \mathbb{C})$ and $h : A^\dagger = A$. Define:

$$\text{ofHermitian}(A, h) : \text{ComplexHermitianMatrix}(n) := \langle A, h \rangle$$

Lemma 10.6.4 (ofHermitian_val). For $A : \text{Matrix}(n, n, \mathbb{C})$ and $h : A^\dagger = A$:

$$(\text{ofHermitian}(A, h)).\text{val} = A$$

Theorem 10.6.5 (val_isHermitian). Let $A : \text{ComplexHermitianMatrix}(n)$. Then $A.\text{val}$ is Hermitian.

10.6.2 Diagonal Entries are Real

Theorem 10.6.6 (diag_re_eq_self). Let $A : \text{ComplexHermitianMatrix}(n)$ and $i : n$. Then:

$$\Re(A_{ii}) = A_{ii}$$

That is, diagonal entries are real.

Theorem 10.6.7 (diag_re_eq_diag). Let $A : \text{ComplexHermitianMatrix}(n)$. Then:

$$\lambda i. \Re(\text{diag}(A)_i) = \text{diag}(A)$$

Theorem 10.6.8 (apply_conj). Let $A : \text{ComplexHermitianMatrix}(n)$ and $i, j : n$. Then:

$$\overline{A_{ji}} = A_{ij}$$

10.6.3 Instances

Proposition 10.6.9. $\text{ComplexHermitianMatrix}(n)$ is an instance of JordanAlgebra .

Proposition 10.6.10. $\text{ComplexHermitianMatrix}(n)$ is an instance of $\text{FormallyRealJordan}$.

10.6.4 Special Matrices

Definition 10.6.11 (one). The identity matrix in $\text{ComplexHermitianMatrix}(n)$:

$$\text{one} := \text{HermitianMatrix.one}$$

Definition 10.6.12 (zero). The zero matrix in $\text{ComplexHermitianMatrix}(n)$:

$$\text{zero} := \langle 0, \cdot \rangle$$

Definition 10.6.13 (realDiagonal). Let $d : n \rightarrow \mathbb{R}$. The **real diagonal matrix** is:

$$\text{realDiagonal}(d) : \text{ComplexHermitianMatrix}(n) := \text{diagonal}(\lambda i. d_i)$$

where entries are embedded as complex numbers. This is Hermitian since real numbers are self-conjugate.

Lemma 10.6.14 (realDiagonal_val). For $d : n \rightarrow \mathbb{R}$:

$$(\text{realDiagonal}(d)).\text{val} = \text{diagonal}(\lambda i. d_i)$$

Definition 10.6.15 (diagonal). Let $d : n \rightarrow \mathbb{C}$ with $\forall i, \overline{d_i} = d_i$. The **diagonal matrix** is:

$$\text{diagonal}(d, hd) : \text{ComplexHermitianMatrix}(n) := \langle \text{Matrix.diagonal}(d), \cdot \rangle$$

10.6.5 Trace Inner Product

Definition 10.6.16 (traceInner). The **trace inner product** for complex Hermitian matrices A, B :

$$\text{tracelInner}(A, B) : \mathbb{C} := \text{HermitianMatrix.tracelInner}(A, B)$$

Definition 10.6.17 (traceInnerReal). The **real trace inner product** for complex Hermitian matrices A, B :

$$\text{tracelInnerReal}(A, B) : \mathbb{R} := \text{HermitianMatrix.tracelInnerReal}(A, B)$$

For Hermitian matrices, $\text{Tr}(AB)$ is always real, so this extracts the real part.

Theorem 10.6.18 (traceInner_comm). For $A, B : \text{ComplexHermitianMatrix}(n)$:

$$\text{tracelInner}(A, B) = \text{tracelInner}(B, A)$$

Theorem 10.6.19 (traceInnerReal_comm). For $A, B : \text{ComplexHermitianMatrix}(n)$:

$$\text{tracelInnerReal}(A, B) = \text{tracelInnerReal}(B, A)$$

Theorem 10.6.20 (traceInnerReal_self_nonneg). For $A : \text{ComplexHermitianMatrix}(n)$:

$$0 \leq \text{tracelInnerReal}(A, A)$$

Chapter 11

Quaternionic Jordan Algebras

11.1 Quaternion: Jordan Product

11.1.1 Jordan Product Definition

Definition 11.1.1 (jordanMul). The **Jordan product** on quaternionic matrices is defined as follows. For matrices $A, B : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$, define:

$$\text{jordanMul}(A, B) := 2^{-1} \cdot (A \cdot B + B \cdot A)$$

where 2^{-1} denotes real scalar multiplication.

Lemma 11.1.2 (jordanMul_def). For all $A, B : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(A, B) = 2^{-1} \cdot (A \cdot B + B \cdot A)$$

11.1.2 Jordan Product Properties

Theorem 11.1.3 (jordanMul_comm). The Jordan product is commutative. For all $A, B : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(A, B) = \text{jordanMul}(B, A)$$

Theorem 11.1.4 (jordanMul_one). For all $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(A, 1) = A$$

Theorem 11.1.5 (one_jordanMul). For all $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(1, A) = A$$

Theorem 11.1.6 (jordanMul_zero). For all $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(A, 0) = 0$$

Theorem 11.1.7 (zero_jordanMul). For all $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(0, A) = 0$$

Theorem 11.1.8 (jordanMul_self). The Jordan square equals the matrix square. For all $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$\text{jordanMul}(A, A) = A \cdot A$$

11.1.3 Hermitian Preservation

Theorem 11.1.9 (`hermitian_jordanMul`). Quaternionic Hermitian matrices are closed under the Jordan product. If $A, B : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$ are Hermitian, then $\text{jordanMul}(A, B)$ is Hermitian.

11.1.4 Jordan Product on QuaternionHermitianMatrix

Definition 11.1.10 (`jmul`). The **Jordan product** on quaternionic Hermitian matrices is defined as follows. For $A, B : \text{QuaternionHermitianMatrix}(n)$, define:

$$\text{jmul}(A, B) := \langle \text{jordanMul}(A.\text{val}, B.\text{val}), _ \rangle$$

where the proof witness follows from `hermitian_jordanMul`.

Lemma 11.1.11 (`jmul_val`). For all $A, B : \text{QuaternionHermitianMatrix}(n)$:

$$(\text{jmul}(A, B)).\text{val} = \text{jordanMul}(A.\text{val}, B.\text{val})$$

Theorem 11.1.12 (`jmul_comm`). The Jordan product on $\text{QuaternionHermitianMatrix}(n)$ is commutative. For all A, B :

$$\text{jmul}(A, B) = \text{jmul}(B, A)$$

Theorem 11.1.13 (`jmul_one`). For all $A : \text{QuaternionHermitianMatrix}(n)$:

$$\text{jmul}(A, \text{one}) = A$$

Theorem 11.1.14 (`one_jmul`). For all $A : \text{QuaternionHermitianMatrix}(n)$:

$$\text{jmul}(\text{one}, A) = A$$

11.2 Quaternion: Hermitian Matrices

11.2.1 CharZero Instance

Proposition 11.2.1 (`Quaternion.instCharZero`). $\mathbb{H}_{\mathbb{R}}$ (quaternions over $\mathbb{R}$) has characteristic zero.

11.2.2 StarModule Instance

Proposition 11.2.2 (`Quaternion.instStarModule`). $\mathbb{H}_{\mathbb{R}}$ forms a **StarModule** over $\mathbb{R}$. That is, real scalar multiplication commutes with the star operation:

$$\star(r \cdot q) = r \cdot \star(q)$$

for all $r \in \mathbb{R}$ and $q \in \mathbb{H}_{\mathbb{R}}$.

11.2.3 Scalar-Star Commutativity

Theorem 11.2.3 (`Quaternion.smul_star_comm`). For $r \in \mathbb{R}$ and $q \in \mathbb{H}_{\mathbb{R}}$:

$$\star(r \cdot q) = r \cdot \star(q)$$

11.2.4 Quaternionic Hermitian Matrix Type

Definition 11.2.4 (QuaternionHermitianMatrix). For a finite type n , define

$$\text{QuaternionHermitianMatrix}(n) := \text{HermitianMatrix}(n, \mathbb{H}_{\mathbb{R}})$$

These are matrices A satisfying $A^H = A$ where $(\cdot)^H$ denotes conjugate transpose using quaternion conjugation.

11.2.5 Basic Properties

Theorem 11.2.5 (conjTranspose_eq). Let A be a quaternionic Hermitian matrix. Then $A^H = A$.

Definition 11.2.6 (ofHermitian). Let $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$ with $h : A^H = A$. Define

$$\text{ofHermitian}(A, h) : \text{QuaternionHermitianMatrix}(n) := \langle A, h \rangle$$

Lemma 11.2.7 (ofHermitian_val). For $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$ with $h : A^H = A$:

$$\text{ofHermitian}(A, h).\text{val} = A$$

Theorem 11.2.8 (val_isHermitian). For $A : \text{QuaternionHermitianMatrix}(n)$, the underlying matrix $A.\text{val}$ is Hermitian.

11.2.6 Diagonal Entries are Real

Theorem 11.2.9 (diag_self_conj). Let $A : \text{QuaternionHermitianMatrix}(n)$ and $i \in n$. Then

$$\star(A_{ii}) = A_{ii}$$

That is, diagonal entries are self-conjugate (real quaternions).

11.2.7 Conjugate Transpose Properties

Theorem 11.2.10 (conjTranspose_smul_real). For $r \in \mathbb{R}$ and $A : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$:

$$(r \cdot A)^H = r \cdot A^H$$

Theorem 11.2.11 (conjTranspose_mul_hermitian). Let $A, B : \text{Matrix}(n, n, \mathbb{H}_{\mathbb{R}})$ be Hermitian. Then

$$(A \cdot B)^H = B \cdot A$$

11.2.8 Special Matrices

Definition 11.2.12 (one). The identity matrix is a quaternionic Hermitian matrix:

$$\text{one} : \text{QuaternionHermitianMatrix}(n) := \langle 1, \text{isHermitian_one} \rangle$$

Definition 11.2.13 (zero). The zero matrix is a quaternionic Hermitian matrix:

$$\text{zero} : \text{QuaternionHermitianMatrix}(n) := \langle 0, \text{trivial} \rangle$$

Definition 11.2.14 (realDiagonal). Let $d : n \rightarrow \mathbb{R}$. Define the real diagonal matrix:

$$\text{realDiagonal}(d) : \text{QuaternionHermitianMatrix}(n) := \text{diag}(d(i))_{i \in n}$$

where each $d(i) \in \mathbb{R}$ is embedded as a real quaternion.

Lemma 11.2.15 (realDiagonal_val). For $d : n \rightarrow \mathbb{R}$:

$$\text{realDiagonal}(d).\text{val} = \text{diag}((d(i) : \mathbb{H}_{\mathbb{R}}))_{i \in n}$$

11.3 Quaternion: Instance

11.3.1 Additivity at Matrix Level

Theorem 11.3.1 (jordanMul_add). For all matrices $A, B, C \in M_n(\mathbb{H})$:

$$A \bullet (B + C) = A \bullet B + A \bullet C$$

where $\bullet$ denotes the Jordan product.

Theorem 11.3.2 (add_jordanMul). For all matrices $A, B, C \in M_n(\mathbb{H})$:

$$(A + B) \bullet C = A \bullet C + B \bullet C$$

11.3.2 Scalar Multiplication at Matrix Level

Theorem 11.3.3 (jordanMul_smul). For all $r \in \mathbb{R}$ and matrices $A, B \in M_n(\mathbb{H})$:

$$(r \cdot A) \bullet B = r \cdot (A \bullet B)$$

Theorem 11.3.4 (smul_jordanMul). For all $r \in \mathbb{R}$ and matrices $A, B \in M_n(\mathbb{H})$:

$$A \bullet (r \cdot B) = r \cdot (A \bullet B)$$

11.3.3 Jordan Identity at Matrix Level

Theorem 11.3.5 (jordanMul_jordan_identity). For all matrices $A, B \in M_n(\mathbb{H})$:

$$(A \bullet B) \bullet (A \bullet A) = A \bullet (B \bullet (A \bullet A))$$

11.3.4 Lifted Properties for Quaternionic Hermitian Matrices

Theorem 11.3.6 (jmul_add). For all quaternionic Hermitian matrices $A, B, C \in \text{QuaternionHermitianMatrix}(n)$:

$$A \circ (B + C) = A \circ B + A \circ C$$

Theorem 11.3.7 (add_jmul). For all quaternionic Hermitian matrices $A, B, C \in \text{QuaternionHermitianMatrix}(n)$:

$$(A + B) \circ C = A \circ C + B \circ C$$

Theorem 11.3.8 (jmul_smul). For all $r \in \mathbb{R}$ and quaternionic Hermitian matrices $A, B \in \text{QuaternionHermitianMatrix}(n)$:

$$(r \cdot A) \circ B = r \cdot (A \circ B)$$

Theorem 11.3.9 (smul_jmul). For all $r \in \mathbb{R}$ and quaternionic Hermitian matrices $A, B \in \text{QuaternionHermitianMatrix}(n)$:

$$A \circ (r \cdot B) = r \cdot (A \circ B)$$

11.3.5 Jordan Identity

Theorem 11.3.10 (jordan_identity). For all quaternionic Hermitian matrices $A, B \in \text{QuaternionHermitianMatrix}(n)$:

$$(A \circ B) \circ (A \circ A) = A \circ (B \circ (A \circ A))$$

11.3.6 JordanAlgebra Instance

Proposition 11.3.11 (moduleQuaternionHermitianMatrix). $\text{QuaternionHermitianMatrix}(n)$ is an instance of $\text{Module } \mathbb{R}$.

Proposition 11.3.12 (jordanAlgebraQuaternionHermitianMatrix). $\text{QuaternionHermitianMatrix}(n)$ is an instance of JordanAlgebra , with:

- $\text{jmul} := \text{QuaternionHermitianMatrix.jmul}$
- $\text{jmul_comm} := \text{QuaternionHermitianMatrix.jmul_comm}$
- $\text{jordan_identity} := \text{QuaternionHermitianMatrix.jordan_identity}$
- $\text{jone} := \text{QuaternionHermitianMatrix.one}$
- $\text{jone_jmul} := \text{QuaternionHermitianMatrix.one_jmul}$
- $\text{jmul_add} := \text{QuaternionHermitianMatrix.jmul_add}$
- $\text{jmul_smul} := \text{QuaternionHermitianMatrix.jmul_smul}$

11.4 Quaternion: Formally Real

Theorem 11.4.1 (quaternion_coe_injective). The coercion $(\cdot) : \mathbb{R} \rightarrow \mathbb{H}$ is injective.

Lemma 11.4.2 (quaternion_sum_coe_eq_coe_sum). For any function $f : n \rightarrow \mathbb{R}$:

$$\sum_{k \in n} (f(k) : \mathbb{H}) = \left(\sum_{k \in n} f(k) \right) : \mathbb{H}$$

Theorem 11.4.3 (diag_mul_self_hermitian). Let $A : M_n(\mathbb{H})$ be a Hermitian matrix (i.e., $A^\dagger = A$). For any $i \in n$:

$$(A \cdot A)_{ii} = \sum_{j \in n} \|A_{ij}\|^2$$

where $\|\cdot\|^2$ denotes the quaternion norm squared.

Theorem 11.4.4 (sum_normSq_eq_zero_iff). Let $A : M_n(\mathbb{H})$ and $i \in n$. Then:

$$\sum_{j \in n} \|A_{ij}\|^2 = 0 \iff \forall j, A_{ij} = 0$$

Theorem 11.4.5 (matrix_sq_eq_zero_imp_zero). Let $A : M_n(\mathbb{H})$ be Hermitian. If $A \cdot A = 0$, then $A = 0$.

Theorem 11.4.6 (jsq_val_eq_mul_self). For a quaternionic Hermitian matrix A , the Jordan square equals the matrix square:

$$A^2.\text{val} = A.\text{val} \cdot A.\text{val}$$

Theorem 11.4.7 (sq_eq_zero_imp_zero). Let A be a quaternionic Hermitian matrix. If $A^2 = 0$ in the Jordan algebra, then $A = 0$.

Definition 11.4.8 (totalNormSq). For a family of quaternionic Hermitian matrices $(A_k)_{k \in \text{Fin}(m)}$ and diagonal index $i \in n$:

$$\text{totalNormSq}(A, i) := \sum_k \sum_j \|(A_k)_{ij}\|^2$$

Lemma 11.4.9 (totalNormSq_nonneg). For any family (A_k) and index i :

$$0 \leq \text{totalNormSq}(A, i)$$

Theorem 11.4.10 (sum_jsq_diag). For a family $(A_k)_{k \in \text{Fin}(m)}$ of quaternionic Hermitian matrices and index $i \in n$:

$$\left(\sum_k A_k^2 \right)_{ii} = \sum_k (A_k \cdot A_k)_{ii}$$

Theorem 11.4.11 (sum_jsq_diag_eq_totalNormSq). For a family $(A_k)_{k \in \text{Fin}(m)}$ of quaternionic Hermitian matrices and index $i \in n$:

$$\left(\sum_k A_k^2 \right)_{ii} = \text{totalNormSq}(A, i)$$

Theorem 11.4.12 (sum_jsq_zero_imp_totalNormSq_zero). Let $(A_k)_{k \in \text{Fin}(m)}$ be a family of quaternionic Hermitian matrices. If $\sum_k A_k^2 = 0$, then for all $i \in n$:

$$\text{totalNormSq}(A, i) = 0$$

Theorem 11.4.13 (sum_sq_eq_zero_imp_all_zero). Let $(A_k)_{k \in \text{Fin}(m)}$ be a family of quaternionic Hermitian matrices. If $\sum_k A_k^2 = 0$, then $A_k = 0$ for all k .

Proposition 11.4.14 (formallyReal). The quaternionic Hermitian matrices $\text{QuaternionHermitianMatrix}(n)$ form a formally real Jordan algebra.

Chapter 12

Spin Factors

12.1 Spin Factor: Definition

Definition 12.1.1 (SpinFactor). The **spin factor** V_n is defined as:

$$\text{SpinFactor}(n) := \mathbb{R} \times \mathbb{R}^n$$

where $\mathbb{R}^n$ is the n -dimensional Euclidean space.

Proposition 12.1.2. $\text{SpinFactor}(n)$ is an instance of `Inhabited`, with default element $(0, 0)$.

12.1.1 Jordan Product

Definition 12.1.3 (jordanMul). The **Jordan product** on spin factors is defined by:

$$(\alpha, v) \circ (\beta, w) := (\alpha\beta + \langle v, w \rangle, \alpha w + \beta v)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product on $\mathbb{R}^n$.

Definition 12.1.4 (one). The **identity element** of the spin factor is:

$$\mathbf{1} := (1, 0)$$

12.1.2 Component Accessors

Definition 12.1.5 (scalar). The **scalar component** of a spin factor element $x = (\alpha, v)$ is:

$$\text{scalar}(x) := \alpha = x_1$$

Definition 12.1.6 (vector). The **vector component** of a spin factor element $x = (\alpha, v)$ is:

$$\text{vector}(x) := v = x_2$$

12.1.3 Basic Properties

Theorem 12.1.7 (jordanMul_comm). The Jordan product on spin factors is commutative:

$$\forall x, y \in V_n : x \circ y = y \circ x$$

Theorem 12.1.8 (one_jordanMul). The identity element is a left identity:

$$\forall x \in V_n : \mathbf{1} \circ x = x$$

Theorem 12.1.9 (jordanMul_one). The identity element is a right identity:

$$\forall x \in V_n : x \circ \mathbf{1} = x$$

Theorem 12.1.10 (jordanMul_add). The Jordan product distributes over addition:

$$\forall x, y, z \in V_n : x \circ (y + z) = (x \circ y) + (x \circ z)$$

Theorem 12.1.11 (jordanMul_smul). The Jordan product is compatible with scalar multiplication:

$$\forall r \in \mathbb{R}, \forall x, y \in V_n : (r \cdot x) \circ y = r \cdot (x \circ y)$$

12.1.4 Jordan Square

Definition 12.1.12 (sq). The **square** of a spin factor element is:

$$\text{sq}(x) := x \circ x$$

Lemma 12.1.13 (sq_def). For all $x \in V_n$:

$$\text{sq}(x) = x \circ x$$

Theorem 12.1.14 (sq_scalar). The scalar component of the square is:

$$(\text{sq}(x))_1 = x_1^2 + \langle x_2, x_2 \rangle$$

Theorem 12.1.15 (sq_vector). The vector component of the square is:

$$(\text{sq}(x))_2 = (2x_1) \cdot x_2$$

12.1.5 Jordan Identity

Theorem 12.1.16 (jordan_identity). The spin factor satisfies the Jordan identity:

$$\forall x, y \in V_n : (x \circ y) \circ \text{sq}(x) = x \circ (y \circ \text{sq}(x))$$

12.1.6 Jordan Algebra Instance

Proposition 12.1.17 (jordanAlgebra). $\text{SpinFactor}(n)$ is an instance of JordanAlgebra , with:

- $\text{jmul} := \text{jordanMul}$
- $\text{jmul_comm} := \text{jordanMul_comm}$
- $\text{jordan_identity} := \text{jordan_identity}$
- $\text{jone} := \text{one}$
- $\text{jone_jmul} := \text{one_jordanMul}$
- $\text{jmul_add} := \text{jordanMul_add}$
- $\text{jmul_smul} := \text{jordanMul_smul}$

Definition 12.1.18 (Notation). The infix notation for the Jordan product on spin factors:

$$x \circ_v y := \text{jordanMul}(x, y)$$

12.2 Spin Factor: Formally Real

12.2.1 Auxiliary Lemmas

Lemma 12.2.1 (add_eq_zero_of_sq_add_nonneg). Let $a, b \in \mathbb{R}$ with $a \geq 0$ and $b \geq 0$. If $a + b = 0$, then $a = 0$ and $b = 0$.

12.2.2 Key Lemmas

Theorem 12.2.2 (sq_eq_zero_imp_scalar_zero). Let $x \in \text{SpinFactor}_n$. If $x^2 = 0$, then $x_1 = 0$.

Theorem 12.2.3 (sq_eq_zero_imp_vector_zero). Let $x \in \text{SpinFactor}_n$. If $x^2 = 0$, then $x_2 = 0$.

Theorem 12.2.4 (sq_eq_zero_imp_zero). Let $x \in \text{SpinFactor}_n$. If $x^2 = 0$, then $x = 0$.

12.2.3 Connection to Jordan Algebra

Lemma 12.2.5 (sq_eq_jsq). For all $x \in \text{SpinFactor}_n$:

$$\text{sq}(x) = x^2$$

Theorem 12.2.6 (jsq_scalar). For $x \in \text{SpinFactor}_n$, the scalar component of the Jordan square is:

$$(x^2)_1 = x_1^2 + \langle x_2, x_2 \rangle$$

Theorem 12.2.7 (sum_jsq_scalar). For a family $a : \text{Fin}(m) \rightarrow \text{SpinFactor}_n$:

$$\left(\sum_i a_i^2 \right)_1 = \sum_i ((a_i)_1^2 + \langle (a_i)_2, (a_i)_2 \rangle)$$

Theorem 12.2.8 (jsq_scalar_nonneg). For all $x \in \text{SpinFactor}_n$:

$$(x^2)_1 \geq 0$$

Theorem 12.2.9 (sum_jsq_eq_zero_imp_scalar_zero). Let $a : \text{Fin}(m) \rightarrow \text{SpinFactor}_n$. If $\sum_i a_i^2 = 0$, then for all $i \in \text{Fin}(m)$:

$$(a_i)_1^2 + \langle (a_i)_2, (a_i)_2 \rangle = 0$$

Theorem 12.2.10 (scalar_term_zero_imp_zero). Let $x \in \text{SpinFactor}_n$. If $x_1^2 + \langle x_2, x_2 \rangle = 0$, then $x = 0$.

12.2.4 Formally Real Instance

Proposition 12.2.11 (formallyReal). SpinFactor_n is an instance of `FormallyRealJordan`.